

The exponential function, its properties and graph

The variable moves into the index — and everything about the shape follows from whether the base is bigger or smaller than 1.

LESSON 49–50

2 × 40 MIN

ALGEBRA 10, §3.1

P2 · 2.1

LESSON CLOCK

10'

22'

20'

22'

16'

10'

The definition	10 min
Two shapes	22 min
The properties	20 min
Growth and decay	22 min
Practice	16 min
Homework	10 min

LEARNING OBJECTIVES

- State the definition and conditions on the base of an exponential function.
- Sketch $y = a^x$ for $a > 1$ and for $0 < a < 1$.
- State the domain, range, monotonicity and asymptote.
- Model exponential growth and decay.

TEXTBOOK

National	pp. 185–196
Cambridge	Pure Mathematics 2 & 3, pp. 26–32
Workbook	P2 Exercise 2A

01

Explanation

The definition

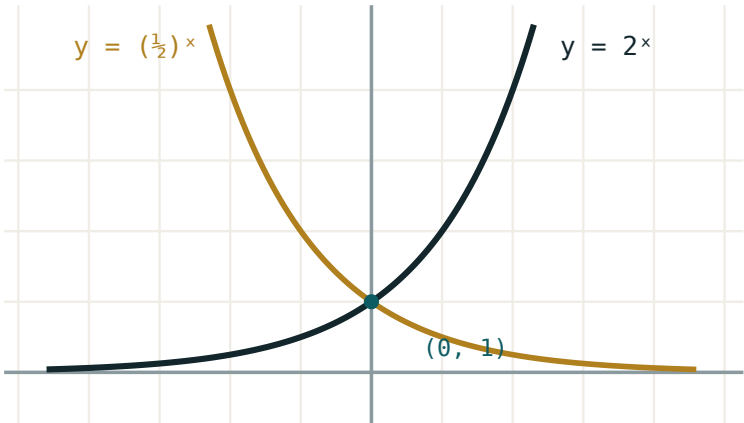
$$y = a^x, a > 0, a \neq 1$$

WHY THE BASE IS RESTRICTED

$a > 0$: otherwise $(-4)^{(1/2)}$ has no real value, and the function has holes. $a \neq 1$: $1^x = 1$ for every x — a constant, not an exponential.

The variable is in the **index**. That is the whole difference from a power function $y = x^n$, and it changes everything: 2^x eventually overtakes x^{100} .

Two shapes



both pass through $(0, 1)$; the x -axis is an asymptote

Growth when the base exceeds 1, decay when it is between 0 and 1. Both pass through $(0, 1)$.

PROPERTY	$A > 1$	$0 < A < 1$
domain	$\mathbb{R}$	$\mathbb{R}$
range	$y > 0$	$y > 0$
behaviour	increasing	decreasing
passes through	$(0, 1)$	$(0, 1)$
asymptote	$y = 0$	$y = 0$
as $x \rightarrow +\infty$	$y \rightarrow \infty$	$y \rightarrow 0$

THE GRAPH NEVER TOUCHES THE x -AXIS

a^x is positive for every real x . The range is $y > 0$, not $y \geq 0$ — and that is why every exponential equation can be divided by a^x safely.

The properties, and why they matter

Every one of these is used in the next four lessons:

- 1. **Monotonic**, so **one-to-one** — hence $a^m = a^n \Rightarrow m = n$, the whole method of solving exponential equations.
- 2. **Positive**, so $a^x = -3$ has no solution and $a^x > 0$ is always true.
- 3. **Increasing for $a > 1$** , so $a^m > a^n \Leftrightarrow m > n$; but for $0 < a < 1$ the inequality **reverses**.
- 4. **Invertible**, and its inverse is the logarithm of Lesson 57–58.

$$a^{(m+n)} = a^m a^n \cdot a^{(m-n)} = \frac{a^m}{a^n} \cdot (a^m)^n = a^{(mn)} \cdot a^0 = 1$$

Growth and decay

$$N(t) = N_0 \cdot a^t \quad a > 1 \text{ growth, } 0 < a < 1 \text{ decay}$$

SITUATION	MODEL	MEANING OF THE BASE
bacteria doubling hourly	$N = N_0 \cdot 2^t$	doubling time 1 hour
a 12% annual rise	$N = N_0 \cdot 1.12^t$	$1 + \frac{12}{100}$
a 7% annual fall	$N = N_0 \cdot 0.93^t$	$1 - \frac{7}{100}$
half-life h	$N = N_0 \cdot (\frac{1}{2})^{(t/h)}$	halving every h

Example. Uzbekistan's population grows about 1.7% a year. From 36 million, after 10 years: $36 \times 1.017^{10} \approx 42.6$ million.

02

Worked examples

EXAMPLE 1 Sketch $y = 3^x$ and state its domain, range and asymptote.

1

$a = 3 > 1$, so increasing.

2

Through $(0, 1)$ and $(1, 3)$.

3

Domain $\mathbb{R}$, range $y > 0$.

4

Asymptote $y = 0$.

ANSWER

Increasing; domain $\mathbb{R}$, range $y > 0$, asymptote $y = 0$

EXAMPLE 2 A colony of 500 bacteria doubles every 3 hours. Find the number after 12 hours.

$$① \quad N = 500 \times 2^{(t/3)}$$

$$② \quad t = 12 \Rightarrow 2^4 = 16$$

$$③ \quad 500 \times 16 = 8000$$

ANSWER

8000

EXAMPLE 3 A radioactive sample has half-life 5 days. What fraction remains after 20 days?

$$① \quad 20 \div 5 = 4 \text{ half-lives.}$$

$$② \quad \left(\frac{1}{2}\right)^4$$

$$③ \quad = \frac{1}{16}$$

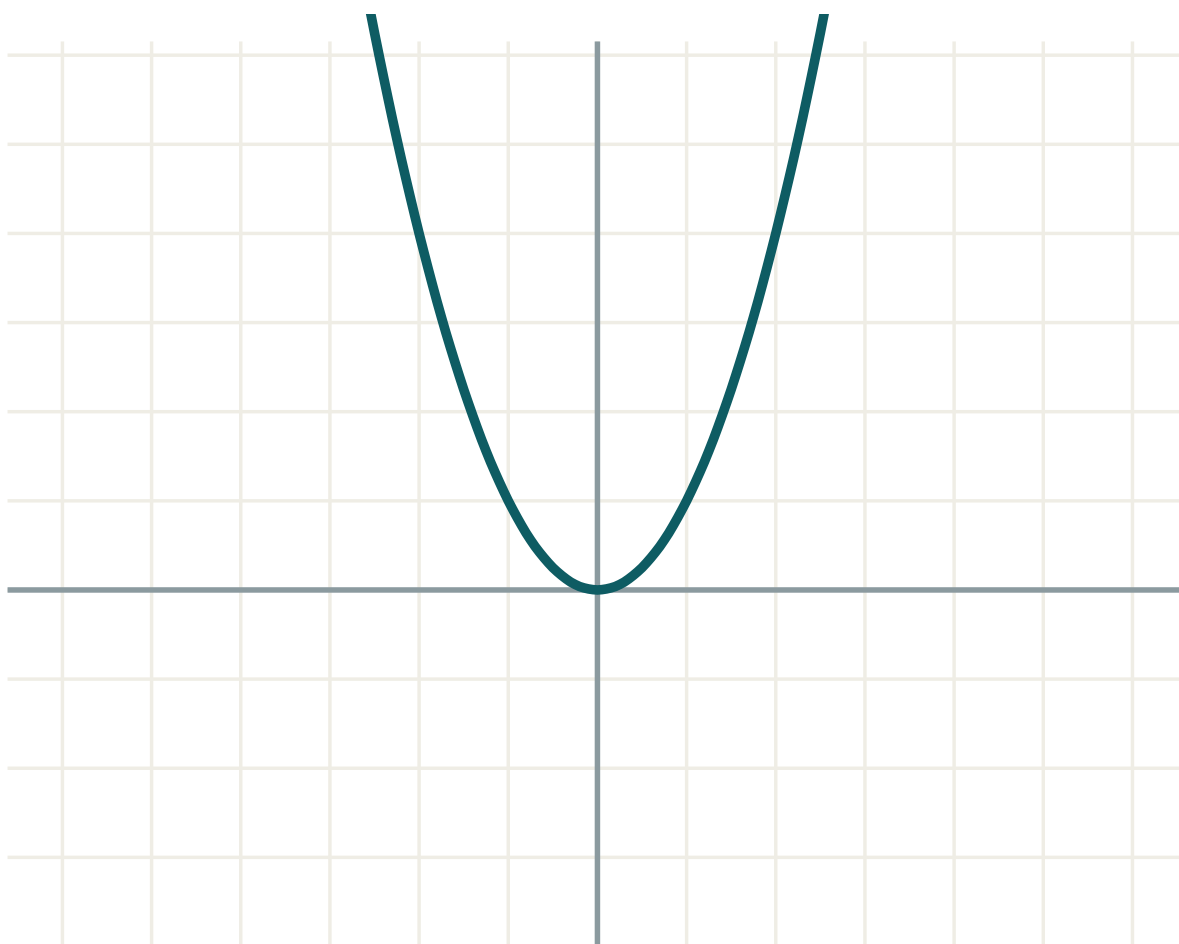
ANSWER

$\frac{1}{16}$ — about 6.25%

03 Interactive model

Change the base through 1 and watch the graph flip from growth to decay.

The exponential curve



a 1

b 0

c 0

equation $y = f(x)$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Exponential function	Ko'rsatkichli funksiya	Показательная функция
Base	Asos	Основание
Index (exponent)	Ko'rsatkich	Показатель
Growth	O'sish	Рост
Decay	Kamayish	Убывание
Horizontal asymptote	Gorizontal asimptota	Горизонтальная асимптота
Monotonic	Monoton	Монотонная
Doubling time	Ikkilanish vaqti	Время удвоения
Half-life	Yarim yemirilish davri	Период полураспада

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The range of $y = a^x$ is:

$y = a^x$ always passes through:

$y = 0.4^x$ is:

The base of an exponential must satisfy:

$$a > 0$$

$$a \neq 1$$

both

neither

A 5% annual fall gives base:

$$0.5$$

$$0.95$$

$$1.05$$

$$5$$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Domain of $y = 2^x$

ANSWER $\mathbb{R}$

2. Range of $y = 2^x$

ANSWER $y > 0$

3. Is $y = 5^x$ increasing?

ANSWER yes

4. Is $y = 0.3^x$ increasing?

ANSWER no

5. 3^0

ANSWER 1

6. 2^5

ANSWER 32

7. $(\frac{1}{2})^3$

ANSWER $\frac{1}{8}$

Medium · the standard question

7 PROBLEMS

1. Base for a 12% annual rise

ANSWER 1.12

2. Base for a 7% annual fall

ANSWER 0.93

3. 500 bacteria doubling every 3 h: number after 12 h

ANSWER 8000

4. Half-life 5 days: fraction left after 20 days

ANSWER $\frac{1}{16}$

5. 36 million growing 1.7% a year: after 10 years

ANSWER ≈ 42.6 million

6. Asymptote of $y = 2^x + 3$

ANSWER $y = 3$

7. Range of $y = 2^x + 3$

ANSWER $y > 3$

Hard · stretch and reason

7 PROBLEMS

1. Sketch $y = 2^{x-1} - 4$: asymptote and intercepts

ANSWER asymptote $y = -4$, x -intercept 3

2. A car worth 200 million so'm falls 15% a year. Value after 4 years

ANSWER ≈ 104.4 million

3. When does $N = 1000 \times 1.05^t$ first exceed 2000?

ANSWER $t \approx 14.2$ years

4. A half-life of 8 days: what percentage remains after 12 days?

ANSWER $\approx 35.4\%$

5. Show 2^x eventually exceeds x^{10}

ANSWER Exponential growth beats any power

6. Solve $2^x = 8$ by inspection

ANSWER $x = 3$

7. Explain why $a^x = -2$ has no solution

ANSWER The range is $y > 0$

07 Homework

Homework — 5 tasks

Every sketch must show the asymptote as a dashed line.

- Sketch $y = 4^x$ and $y = (\frac{1}{4})^x$ on one set of axes, with domain, range and asymptote.
- Sketch $y = 3^x - 9$ and give its asymptote and both intercepts.
- A town of 40 000 grows 2.5% a year. Find its population after 8 years.
- A sample has half-life 6 hours. What fraction remains after 24 hours?
- Explain in three sentences why the base of an exponential function may not be 1.

Exponential equations

Get the same base on both sides, then equate the indices — and know the three tricks for when you cannot.

LESSON 51–52

2 × 40 MIN

ALGEBRA 10, §3.2

P2 · 2.3

LESSON CLOCK

10'

22'

24'

22'

20'

12'

The one-to-one property	10 min
A common base	22 min
Substitution	24 min
Homogeneous equations	22 min
Practice	20 min
Homework	12 min

LEARNING OBJECTIVES

- Solve $a^{f(x)} = a^{g(x)}$ by equating indices.
- Rewrite both sides to a common base.
- Solve equations that become quadratic after a substitution.
- Solve homogeneous equations by dividing through.

TEXTBOOK

National	pp. 197–208
Cambridge	Pure Mathematics 2 & 3, pp. 33–38
Workbook	P2 Exercise 2B

01

Explanation

The method rests on one property

A^x IS ONE-TO-ONE

Because $y = a^x$ is strictly monotonic, $a^m = a^n$ forces $m = n$ (for $a > 0$, $a \neq 1$). That single fact solves most exponential equations.

$$2^{(3x-1)} = 2^{(x+5)} \Rightarrow 3x - 1 = x + 5 \Rightarrow x = 3$$

So the whole task is: **make the bases the same.**

Getting a common base

EQUATION	REWRITE AS	SOLUTION
$2^x = 32$	$2^x = 2^5$	$x = 5$
$3^{(2x)} = 81$	$3^{(2x)} = 3^4$	$x = 2$
$4^x = 8$	$2^{(2x)} = 2^3$	$x = 1.5$
$(\frac{1}{3})^x = 9$	$3^{(-x)} = 3^2$	$x = -2$
$5^{(x^2-3x)} = 1$	$5^{(x^2-3x)} = 5^0$	$x = 0, 3$

EVERY NUMBER IS A POWER OF SOMETHING

$32 = 2^5$, $81 = 3^4$, $8 = 2^3$, $0.25 = 2^{-2}$, $1 = a^0$. Learning the powers of 2, 3, 5 up to a few hundred removes most of the difficulty in this topic.

Substitution

When $a^{(2x)}$ and a^x both appear, set $t = a^x$ — and remember $t > 0$:

$$4^x - 5 \cdot 2^x + 4 = 0, t = 2^x > 0 \Rightarrow t^2 - 5t + 4 = 0$$

$t = 1$ or $t = 4$; both positive, so $2^x = 1$ gives $x = 0$ and $2^x = 4$ gives $x = 2$.

WRITE $T > 0$ WITH THE SUBSTITUTION

A negative t is rejected immediately, because a^x is never negative. That single line saves the whole check at the end — and it is the mark the examiner is looking for.

Note $4^x = (2^2)^x = (2^x)^2 = t^2$: recognising that is the step, not the algebra after it.

Homogeneous equations

When every term has the same total degree in two bases, divide through by one of them:

$$9^x - 4 \cdot 6^x + 3 \cdot 4^x = 0$$

Divide by 4^x (never zero) to get, with $t = (\frac{3}{2})^x > 0$:

$$t^2 - 4t + 3 = 0 \Rightarrow t = 1, 3 \Rightarrow x = 0, x = \log_{(1.5)} 3$$

The pattern to spot: $9^x = (3^2)^x$, $6^x = (2 \cdot 3)^x$, $4^x = (2^2)^x$ — the bases 9, 6, 4 are 3^2 , $3 \cdot 2$, 2^2 , a perfect quadratic pattern.

02 Worked examples

EXAMPLE 1 Solve $2^{x+3} = 32$.

① $32 = 2^5$

② $x + 3 = 5$

③ $x = 2$

ANSWER

$x = 2$

EXAMPLE 2 Solve $9^x - 10 \cdot 3^x + 9 = 0$.

① Let $t = 3^x > 0$; $9^x = t^2$.

② $t^2 - 10t + 9 = 0$

③ $t = 1$ or $t = 9$, both positive.

④ $3^x = 1 \Rightarrow x = 0$; $3^x = 9 \Rightarrow x = 2$.

ANSWER

$x = 0$ and $x = 2$

EXAMPLE 3 Solve $2^{x+1} + 2^{x-1} = 20$.

$$① \quad 2 \cdot 2^x + \frac{2^x}{2} = 20$$

Factor out 2^x .

$$② \quad 2^x(2 + 0.5) = 20$$

$$③ \quad 2^x = 8$$

$$④ \quad x = 3$$

ANSWER

$$x = 3$$

03 Interactive model

Write the powers of 2 and 3 on the board and leave them there for the whole lesson.

Powers and their bases

$$\sqrt[3]{64} = 64^{1/3} = 4$$

base a **64**index n **3**power m **1**check: value³ **64**a¹ **64**

Raising the answer to the power 3 gives back a^1 — that is exactly what “root of degree 3” means.

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Exponential equation	Ko'rsatkichli tenglama	Показательное уравнение
Common base	Umumiy asos	Общее основание
Equating indices	Ko'rsatkichlarni tenglashtirish	Приравнивание показателей
Substitution	Almashtirish	Замена
Homogeneous equation	Bir jinsli tenglama	Однородное уравнение
Auxiliary variable	Yordamchi o'zgaruvchi	Вспомогательная переменная
Positivity condition	Musbatlik sharti	Условие положительности
Rejected root	Rad etilgan ildiz	Отброшенный корень

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

$2^x = 64$ gives:

$a^m = a^n$ implies $m = n$ because a^x is:

For $4^x - 5 \cdot 2^x + 4 = 0$ substitute:

The condition on that t is:

$$t \geq 0$$

$$t > 0$$

$$t \neq 0$$

none

$5^{(x^2-4)} = 1$ gives:

$$x = 4$$

$$x = \pm 2$$

$$x = 0$$

no solution

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Solve $2^x = 16$

ANSWER $x = 4$

2. Solve $3^x = 27$

ANSWER $x = 3$

3. Solve $5^x = 1$

ANSWER $x = 0$

4. Solve $2^{(x+1)} = 8$

ANSWER $x = 2$

5. Solve $10^x = 0.01$

ANSWER $x = -2$

6. Solve $4^x = 2$

ANSWER $x = 0.5$

7. Solve $(\frac{1}{2})^x = 8$

ANSWER $x = -3$

Medium · the standard question

7 PROBLEMS

1. Solve $2^{3x-1} = 2^{x+5}$

ANSWER $x = 3$

2. Solve $9^x = 27$

ANSWER $x = 1.5$

3. Solve $5^{(x^2-3x)} = 1$

ANSWER $x = 0, 3$

4. Solve $4^x - 5 \cdot 2^x + 4 = 0$

ANSWER $x = 0, 2$

5. Solve $9^x - 10 \cdot 3^x + 9 = 0$

ANSWER $x = 0, 2$

6. Solve $2^{(x+1)} + 2^{(x-1)} = 20$

ANSWER $x = 3$

7. Solve $3^{(x+2)} = 9^{(x-1)}$

ANSWER $x = 4$

Hard · stretch and reason

7 PROBLEMS

1. Solve $4^x - 3 \cdot 2^x - 4 = 0$

ANSWER $x = 2$

2. Solve $9^x - 4 \cdot 6^x + 3 \cdot 4^x = 0$

ANSWER $x = 0$ and $x = \log_{1.5} 3$

3. Solve $2^{2x} - 6 \cdot 2^x + 8 = 0$

ANSWER $x = 1, 2$

4. Solve $3^{x+1} + 3^{x-1} = 30$

ANSWER $x = 2$

5. Solve $25^x - 6 \cdot 5^x + 5 = 0$

ANSWER $x = 0, 1$

6. Solve $2^{x^2} = 4^x$

ANSWER $x = 0, 2$

7. Explain why $4^x + 2^x + 1 = 0$ has no solution

ANSWER Every term is positive

07 Homework

Homework — 6 tasks

Write $t > 0$ with every substitution.

1. Solve $3^{2x-1} = 81$ and $8^x = 32$.

2. Solve $2^{x^2-5x} = 1$.

3. Solve $9^x - 4 \cdot 3^x + 3 = 0$.

4. Solve $5^{x+1} + 5^{x-1} = 130$.

5. Solve $4^x - 6 \cdot 2^x + 8 = 0$.

6. Explain in three sentences why the substitution $t = a^x$ must carry the condition $t > 0$.

Exponential inequalities

The same method as the equations, with one extra question that decides the direction of the answer.

LESSON 53–54 2 × 40 MIN ALGEBRA 10, §3.3 P2 · 2.4

LESSON CLOCK

10'	22'	22'	24'	20'	12'
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The one extra question	10 min
Base greater than 1	22 min
Base less than 1	22 min
Substitution	24 min
Practice	20 min
Homework	12 min

LEARNING OBJECTIVES

- Use monotonicity to convert an exponential inequality into one about the indices.
- Reverse the inequality when the base is less than 1.
- Solve inequalities that become quadratic after a substitution.
- Write the solution in interval notation.

TEXTBOOK

National	pp. 209–220
Cambridge	Pure Mathematics 2 & 3, pp. 39–44
Workbook	P2 Exercise 2C

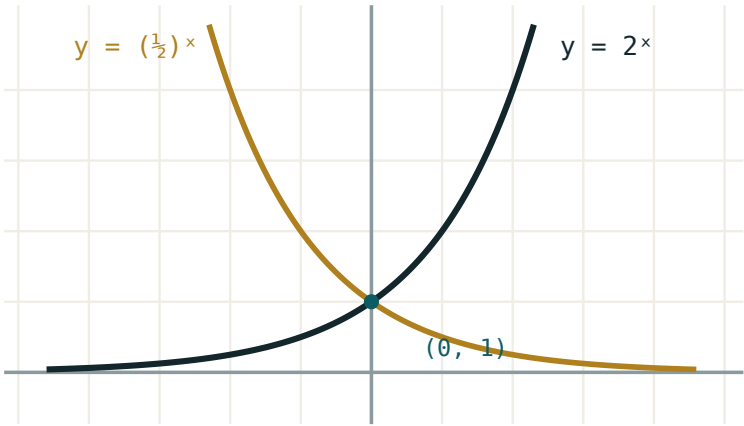
01 Explanation

The one extra question

ASK IT BEFORE ANYTHING ELSE

Is the base bigger or smaller than 1? For $a > 1$ the function increases, so the inequality between the indices points the **same** way. For $0 < a < 1$ it decreases, so the inequality **reverses**.

$$a > 1: a^m > a^n \Leftrightarrow m > n \quad 0 < a < 1: a^m > a^n \Leftrightarrow m < n$$



both pass through (0, 1); the x-axis is an asymptote

*Read it off the picture: the rising curve preserves order,
the falling curve reverses it.*

Base greater than 1

$$2^{x+1} > 8 \Rightarrow 2^{x+1} > 2^3 \Rightarrow x + 1 > 3 \Rightarrow x > 2$$

The sign is untouched. Every step is the technique of the last lesson, with > in place of =.

INEQUALITY	SAME BASE	ANSWER
$3^x > 27$	$3^x > 3^3$	$x > 3$
$2^x \leq 16$	$2^x \leq 2^4$	$x \leq 4$
$5^{2x} < 125$	$5^{2x} < 5^3$	$x < 1.5$
$4^x \geq 1$	$4^x \geq 4^0$	$x \geq 0$

Base less than 1

$$\left(\frac{1}{2}\right)^x > 8 \Rightarrow \left(\frac{1}{2}\right)^x > \left(\frac{1}{2}\right)^{-3} \Rightarrow x < -3$$

THIS IS THE MARK EVERYONE LOSES

The sign flipped, and nothing on the page says so except the base. Circle the base first; write “base < 1, so reverse” on its own line; then flip.

An alternative that avoids the trap: rewrite $\left(\frac{1}{2}\right)^x$ as 2^{-x} so the base is greater than 1, and let the minus sign do the reversing where you can see it.

Substitution

With $t = a^x > 0$ the inequality becomes an ordinary one in t , solved by the sign chart of Quarter II — but the answer must be translated back:

$$4^x - 5 \cdot 2^x + 4 < 0, t = 2^x > 0 \Rightarrow t^2 - 5t + 4 < 0 \Rightarrow 1 < t < 4$$

$$1 < 2^x < 4 \Rightarrow 0 < x < 2$$

TRANSLATE CAREFULLY

$1 < t < 4$ becomes $2^0 < 2^x < 2^2$. Because the base exceeds 1, the order survives and the answer is $0 < x < 2$. With a base under 1 the whole double inequality would reverse.

02 Worked examples

EXAMPLE 1 Solve $3^{2x-1} \geq 27$.

- ① $27 = 3^3$; base $3 > 1$.
Sign unchanged.

② $2x - 1 \geq 3$

③ $x \geq 2$

ANSWER

$$x \geq 2$$

EXAMPLE 2 Solve $(\frac{1}{3})^{(x+2)} > 9$.

① $9 = (\frac{1}{3})^{(-2)}$; base < 1 .
Reverse.

② $x + 2 < -2$

③ $x < -4$

ANSWER

$$x < -4$$

EXAMPLE 3 Solve $9^x - 4 \cdot 3^x + 3 \leq 0$.

① Let $t = 3^x > 0$.
 $t^2 - 4t + 3 \leq 0$

② $(t - 1)(t - 3) \leq 0 \Rightarrow 1 \leq t \leq 3$

③ $3^0 \leq 3^x \leq 3^1$
Base > 1 .

④ $0 \leq x \leq 1$

ANSWER

$$0 \leq x \leq 1$$

03 Interactive model

Draw the rising and falling curves side by side before solving anything.

Reading the solution set

$$x > 2$$

inequality $x > 2$ boundary **2** (**open**) $x = 0$ works? **no** $x = 6$ works? **yes**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Exponential inequality	Ko'rsatkichli tengsizlik	Показательное неравенство
Monotonicity	Monotonlik	Монотонность
Reversing the sign	Ishorani almashtirish	Смена знака неравенства
Base greater than 1	Birdan katta asos	Основание больше единицы
Base between 0 and 1	Birdan kichik asos	Основание меньше единицы
Interval notation	Oraliq belgilanishi	Интервальная запись
Substitution	Almashtirish	Замена
Solution set	Yechimlar to'plami	Множество решений

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

$2^x > 8$ gives:

$(\frac{1}{2})^x > 8$ gives:

The sign reverses when:

$1 < 2^x < 4$ gives:

$$0 < x < 2$$

$$1 < x < 4$$

$$x > 2$$

$$0 < x < 4$$

$4^x \geq 1$ gives:

$$x \geq 1$$

$$x \geq 0$$

$$x > 0$$

$$x \leq 0$$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Solve $2^x > 16$

ANSWER $x > 4$

2. Solve $3^x \leq 9$

ANSWER $x \leq 2$

3. Solve $5^x > 1$

ANSWER $x > 0$

4. Solve $(\frac{1}{2})^x > 4$

ANSWER $x < -2$

5. Solve $(\frac{1}{3})^x < 27$

ANSWER $x > -3$

6. Solve $10^x \geq 100$

ANSWER $x \geq 2$

7. Solve $2^x < 1$

ANSWER $x < 0$

Medium · the standard question

7 PROBLEMS

1. Solve $3^{2x-1} \geq 27$

ANSWER $x \geq 2$

2. Solve $(\frac{1}{3})^{x+2} > 9$

ANSWER $x < -4$

3. Solve $2^{x+1} \leq 32$

ANSWER $x \leq 4$

4. Solve $4^x > 8$

ANSWER $x > 1.5$

5. Solve $9^x \leq 3^{x+2}$

ANSWER $x \leq 2$

6. Solve $(0.2)^x \geq 25$

ANSWER $x \leq -2$

7. Solve $2^{x^2-4} < 1$

ANSWER $-2 < x < 2$

Hard · stretch and reason

7 PROBLEMS

1. Solve $9^x - 4 \cdot 3^x + 3 \leq 0$

ANSWER $0 \leq x \leq 1$

2. Solve $4^x - 5 \cdot 2^x + 4 < 0$

ANSWER $0 < x < 2$

3. Solve $25^x - 6 \cdot 5^x + 5 > 0$

ANSWER $x < 0$ or $x > 1$

4. Solve $2^{x+1} + 2^{x-1} > 20$

ANSWER $x > 3$

5. Solve $(\frac{1}{2})^{x^2} \geq \frac{1}{16}$

ANSWER $-2 \leq x \leq 2$

6. Solve $3^{x+1} < 9^{x-1}$

ANSWER $x > 3$

7. Explain why $2^x > -1$ has solution $\mathbb{R}$

ANSWER Every power of 2 is positive

07 Homework**Homework — 6 tasks**Write “base > 1 ” or “base < 1 ” on its own line before flipping anything.

1. Solve $2^{3x-2} \geq 64$.

2. Solve $(\frac{1}{4})^{x-1} > 16$.

3. Solve $9^x \leq 3^{x+6}$.

4. Solve $4^x - 10 \cdot 2^x + 16 \leq 0$.

5. Solve $3^{x^2-9} < 1$.

6. Explain in three sentences why the base decides the direction of the answer.

Control work 5, and work on the mistakes

The exponential block in one paper — and the drill that fixes the base-under-one trap for good.

LESSON 55–56

2 × 40 MIN

ALGEBRA 10, NAZORAT ISHI 5

P2 · CHAPTER 2 REVIEW

LESSON CLOCK

3

40'

10'

25'

12'

Instructions	3 min
The paper	40 min
Self-mark	10 min
Rewrite	25 min
The base drill	12 min

LEARNING OBJECTIVES

- Apply the exponential methods under time.
- Decide the direction of an inequality from the base without prompting.
- Classify each lost mark as careless, method or knowledge.
- Rewrite every wrong solution correctly.

TEXTBOOK

National	pp. 221–224
Cambridge	Pure Mathematics 2 & 3, pp. 45–46
Workbook	Control paper E

01

Explanation

The paper — 25 marks, 40 minutes

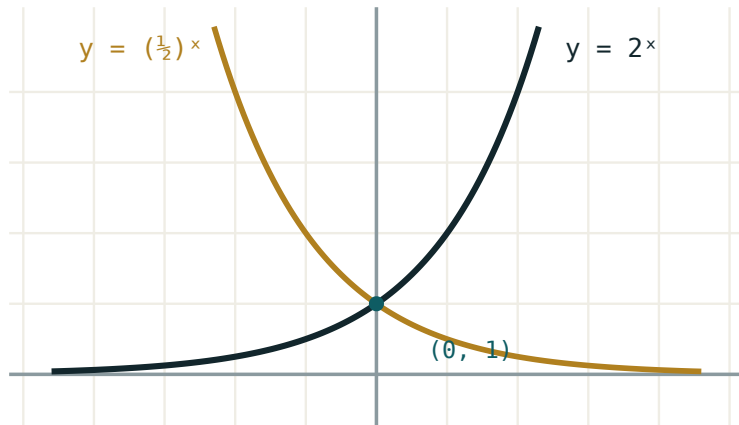
Q	TASK	MARKS	FROM
1	Sketch $y = 2^x - 4$, with asymptote, range and both intercepts	4	L49–50
2	Solve $3^{2x+1} = 27$ and $8^x = 4$	4	L51–52
3	Solve $4^x - 6 \cdot 2^x + 8 = 0$	4	L51–52
4	Solve $2^{x-1} > 32$ and $(\frac{1}{5})^x \geq 25$	5	L53–54
5	Solve $9^x - 10 \cdot 3^x + 9 < 0$	4	L53–54
6	A sample of 80 g has half-life 4 days. Find the mass after 12 days, and when 5 g remain	4	L49–50

Q4 CARRIES THE WHOLE TOPIC IN FIVE MARKS

Two for the first part, three for the second — of which one is for stating that the base is less than 1 and the sign must reverse.

The three errors

ERROR	LOOKS LIKE	KIND
sign not reversed	$(\frac{1}{2})^x > 8 \Rightarrow x > -3$	knowledge
$t > 0$ omitted	a negative t kept and turned into a false root	method
bases not matched	$8^x = 4$ attacked without writing $2^{3x} = 2^2$	method
power miscopied	$27 = 3^3$ written as 3^4	careless



both pass through $(0, 1)$; the x -axis is an asymptote

The picture that settles every direction question in this paper.

The base drill

Twelve inequalities on the board. For each, the class calls out only “**same**” or “**reverse**”, in two seconds:

INEQUALITY	CALL
$3^x > 9$	same
$(\frac{1}{3})^x > 9$	reverse
$0.7^x < 0.49$	reverse
$1.5^x \geq 2.25$	same
$(0.2)^x \leq 5$	reverse

ROW 5 IS THE ONE THEY MISS

$5 = (0.2)^{-1}$. Writing the right-hand side as a power of the **same** base is what makes the comparison possible at all.

02 Worked examples

EXAMPLE 1 Model answer, Q4b: solve $(\frac{1}{5})^x \geq 25$.

- ① Base $\frac{1}{5} < 1$ — the sign will reverse.
One mark for this line.

② $25 = (\frac{1}{5})^{-2}$

③ $x \leq -2$

ANSWER

$$x \leq -2$$

EXAMPLE 2 Model answer, Q5: solve $9^x - 10 \cdot 3^x + 9 < 0$.

- ① Let $t = 3^x > 0$.
 $t^2 - 10t + 9 < 0$

② $(t - 1)(t - 9) < 0 \Rightarrow 1 < t < 9$

③ $3^0 < 3^x < 3^2$
Base > 1 , order preserved.

④ $0 < x < 2$

ANSWER

$$0 < x < 2$$

EXAMPLE 3 A learner wrote $(\frac{1}{2})^x > 8 \Rightarrow x > -3$. Name the mistake.

- ① The base is under 1.
- ② The function decreases, so the inequality reverses.
A knowledge gap.
- ③ Correct: $x < -3$.

ANSWER

Knowledge gap — the reversal rule

03 Interactive model

Run the base drill before the rewrite, and again at the end of the lesson.

Same or reverse?

$3^x > 9:$

$x > 2$

$x < 2$

$x > 9$

$x < 9$

$(\frac{1}{3})^x > 9:$

$x > 2$

$x < -2$

$x > -2$

$x < 2$

$8^x = 4:$

$x = 2$

$x = \frac{2}{3}$

$x = 0.5$

$x = \frac{3}{2}$

$$4^x - 6 \cdot 2^x + 8 = 0:$$

$$x = 1, 2$$

$$x = 2, 4$$

$$x = 0, 3$$

$$x = 2$$

$$2^{(x-1)} > 32:$$

$$x > 5$$

$$x > 6$$

$$x < 6$$

$$x > 4$$

80 g, half-life 4 days, after 12 days:

$$20 \text{ g}$$

$$10 \text{ g}$$

$$40 \text{ g}$$

$$5 \text{ g}$$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Control work	Nazorat ishi	Контрольная работа
Common base	Umumiy asos	Общее основание
Reversing the sign	Ishorani almashtirish	Смена знака
Substitution	Almashtirish	Замена
Careless error	E'tiborsizlik xatosi	Ошибка по невнимательности
Method error	Usul xatosi	Ошибка в методе
Knowledge gap	Bilim bo'shlig'i	Пробел в знаниях
Correction	Tuzatish	Исправление

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The direction of an exponential inequality is decided by:

the index

the base

the constant

the sign of x

A substitution must always carry:

$t \geq 0$

$t > 0$

$t \neq 0$

nothing

$8^x = 4$ is best done by:

guessing

writing both as powers of 2

dividing

squaring

Three half-lives leave:

$$\frac{1}{2}$$

$$\frac{1}{4}$$

$$\frac{1}{8}$$

$$\frac{1}{3}$$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Solve $3^{(2x+1)} = 27$

ANSWER $x = 1$

2. Solve $8^x = 4$

ANSWER $x = \frac{2}{3}$

3. Solve $2^{(x-1)} > 32$

ANSWER $x > 6$

4. Solve $(\frac{1}{5})^x \geq 25$

ANSWER $x \leq -2$

5. Asymptote of $y = 2^x - 4$

ANSWER $y = -4$

6. x -intercept of $y = 2^x - 4$

ANSWER $x = 2$

7. 80 g, half-life 4 days, after 12 days

ANSWER 10 g

Medium · the standard question

7 PROBLEMS

1. Solve $4^x - 6 \cdot 2^x + 8 = 0$

ANSWER $x = 1, 2$

2. Solve $9^x - 10 \cdot 3^x + 9 < 0$

ANSWER $0 < x < 2$

3. Range of $y = 2^x - 4$

ANSWER $y > -4$

4. When do 5 g remain of 80 g with half-life 4 days?

ANSWER 16 days

5. Solve $25^x = 5^{(x+3)}$

ANSWER $x = 3$

6. Solve $(0.2)^x \leq 5$

ANSWER $x \geq -1$

7. Solve $2^{(x^2-1)} = 8$

ANSWER $x = \pm 2$

Hard · stretch and reason

7 PROBLEMS

1. Solve $4^x - 2^{x+3} + 12 = 0$

ANSWER $x = 1$ and $x = \log_2 6$

2. Solve $9^x - 3^{x+2} + 8 \geq 0$

ANSWER $x \leq 0$ or $x \geq \log_3 8$

3. Solve $2^{2x+1} - 5 \cdot 2^x + 2 = 0$

ANSWER $x = -1, 1$

4. Sketch $y = 3^{x-2} + 1$: asymptote and y-intercept

ANSWER $y = 1; \frac{10}{9}$

5. A population trebles every 5 years. From 2000, when does it pass 50 000?

ANSWER ≈ 14.6 years

6. Solve $(\sqrt{2})^x = 8$

ANSWER $x = 6$

7. Solve $3^{2x} - 12 \cdot 3^x + 27 < 0$

ANSWER $1 < x < \log_3 9$ — i.e. $1 < x < 2$

07 Homework

Homework — 4 tasks

Task 1 is the rewrite. State the base rule on every inequality.

1. Rewrite in full every question that lost a mark, with the base rule stated.
2. Five problems from the section your knowledge column was heaviest in.
3. Solve $16^x - 10 \cdot 4^x + 16 = 0$ and $(\frac{1}{7})^{x+1} > 49$.

4. A sample of 120 g has half-life 3 hours. Find the mass after 15 hours and the time until 7.5 g remain.

The logarithm and the logarithmic function

The inverse of the exponential — a new name for an index, and the graph you get by reflecting.

LESSON 57–58

2 × 40 MIN

ALGEBRA 10, §3.4

P2 · 2.2

LESSON CLOCK

12'	20'	24'	20'	18'	6'
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The definition	12 min
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Evaluating	20 min
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The graph	24 min
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Inverse of the exponential	20 min
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Practice	18 min
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Homework	6 min
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LEARNING OBJECTIVES

- State the definition of a logarithm and move between the two forms.
- Evaluate simple logarithms without a calculator.
- Sketch $y = \log_a x$ and state its domain, range and asymptote.
- Recognise the logarithm as the inverse of the exponential.

TEXTBOOK

National	pp. 225–236
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Cambridge	Pure Mathematics 2 & 3, pp. 15–25
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Workbook	P2 Exercise 2D
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01 Explanation

The definition

A LOGARITHM IS AN INDEX

$$\log_a b = c \Leftrightarrow a^c = b \quad (a > 0, a \neq 1, b > 0)$$

In words: $\log_a b$ is the power to which a must be raised to give b .

So $\log_2 8 = 3$ because $2^3 = 8$; $\log_5 25 = 2$ because $5^2 = 25$; and $\log_3 \frac{1}{9} = -2$

because $3^{-2} = \frac{1}{9}$.

THE ARGUMENT MUST BE POSITIVE

$\log_a(-4)$ does not exist: no power of a positive base is negative. That single condition is the first line of every logarithmic equation in the next four lessons.

Evaluating without a calculator

Ask “ a to what power gives b ?” and answer it.

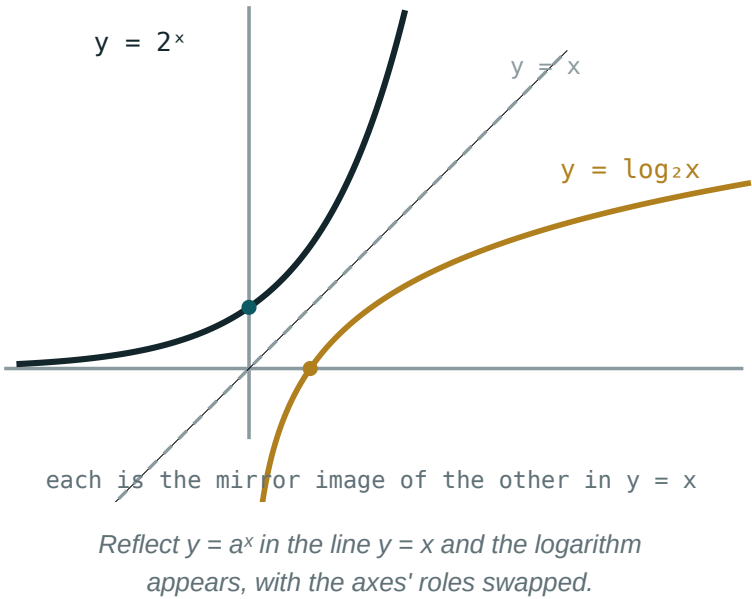
EXPRESSION	QUESTION	VALUE
$\log_2 32$	$2^? = 32$	5
$\log_3 1$	$3^? = 1$	0
$\log_7 7$	$7^? = 7$	1
$\log_4 2$	$4^? = 2$	$\frac{1}{2}$
$\log_2 \frac{1}{8}$	$2^? = \frac{1}{8}$	-3
$\lg 1000$	$10^? = 1000$	3

Two notations are standard: **lg** x means $\log_{10} x$, and **ln** x means $\log_e x$, where $e \approx 2.718$ is the base that makes calculus simplest.

$$\log_a 1 = 0 \cdot \log_a a = 1 \cdot a^{(\log_a b)} = b$$

The last is the **fundamental logarithmic identity**: raising a to the power that gives b gives b . It is the statement that the two operations undo each other.

The graph



PROPERTY	$Y = A^x$	$Y = \text{LOG}_A X$
domain	$\mathbb{R}$	$x > 0$
range	$y > 0$	$\mathbb{R}$
passes through	$(0, 1)$	$(1, 0)$
asymptote	$y = 0$	$x = 0$
$a > 1$	increasing	increasing
$0 < a < 1$	decreasing	decreasing

Every row is the previous table with the two columns swapped — because reflecting in $y = x$ exchanges the axes.

Inverse, and why it matters

$$\log_a(a^x) = x \quad a^{(\log_a x)} = x$$

These are the two halves of “ $f^{-1}f = ff^{-1} = \text{identity}$ ” from Grade 10 Quarter I.

WHAT LOGARITHMS ARE FOR

They take an unknown **out of an index**. $2^x = 10$ cannot be solved by matching bases — but taking $\log_2$ of both sides gives $x = \log_2 10 \approx 3.32$ immediately. That is the whole reason the function exists.

02 Worked examples

EXAMPLE 1 Evaluate $\log_3 81$, $\log_5 \frac{1}{125}$ and $\log_8 2$.

① $3^4 = 81$
4

② $5^{-3} = \frac{1}{125}$
-3

③ $8^{1/3} = 2$
 $\frac{1}{3}$

ANSWER

4, -3, $\frac{1}{3}$

EXAMPLE 2 Solve $\log_2 x = 5$ and $\log_x 49 = 2$.

① $x = 2^5 = 32$

② $x^2 = 49$

③ $x = 7$; the base must be positive and not 1.
 $x = -7$ rejected.

ANSWER

$x = 32$; $x = 7$

EXAMPLE 3 Find the domain of $y = \log_3(x - 4)$.

① The argument must be positive.

② $x - 4 > 0$

③ $x > 4$

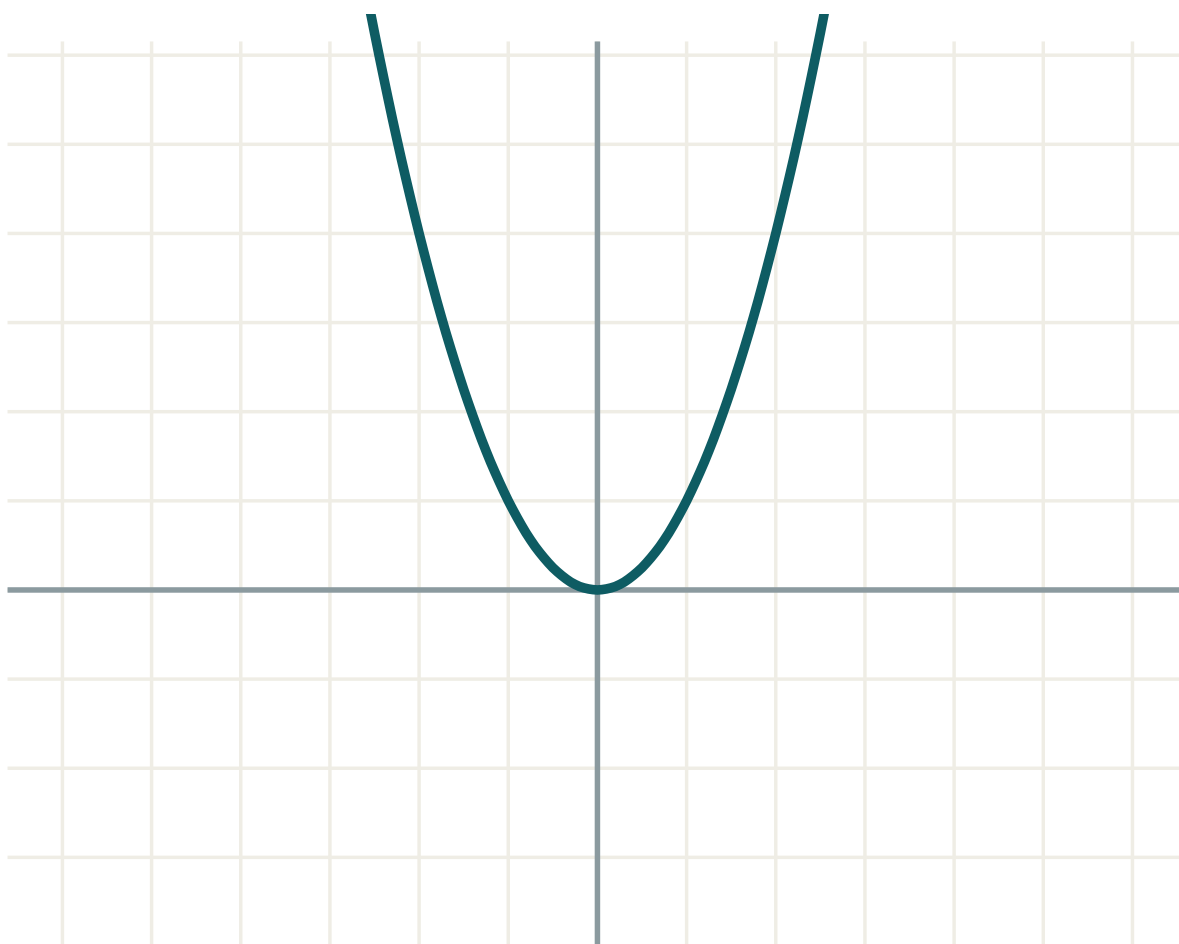
ANSWER

$x > 4$

03 Interactive model

Say each logarithm aloud as a question — “two to what power gives eight?”

The logarithmic curve



a 1

b 0

c 0

equation $y = f(x)$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Logarithm	Logarifm	Логарифм
Base of a logarithm	Logarifm asosi	Основание логарифма
Argument of a logarithm	Logarifm argumenti	Аргумент логарифма
Common logarithm (lg)	O'nli logarifm	Десятичный логарифм
Natural logarithm (ln)	Natural logarifm	Натуральный логарифм
Vertical asymptote	Vertikal asimptota	Вертикальная асимптота
Inverse function	Teskari funksiya	Обратная функция
Logarithmic identity	Logarifmik ayniyat	Основное логарифмическое тождество
Reflection in $y = x$	$y = x$ ga nisbatan akslantirish	Симметрия относительно $y = x$

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

$\log_2 16$ is:

$\log_a 1$ is:

The domain of $y = \log_a x$ is:

$y = \log_a x$ passes through:

$(0, 1)$

$(1, 0)$

$(0, 0)$

$(1, 1)$

$a^{(\log_a 7)}$ equals:

a

7

$\log_a 7$

1

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $\log_2 8$

ANSWER 3

2. $\log_3 9$

ANSWER 2

3. $\log_5 1$

ANSWER 0

4. $\log_7 7$

ANSWER 1

5. $\lg 100$

ANSWER 2

6. $\log_2 \frac{1}{4}$

ANSWER -2

7. $\log_9 3$

ANSWER $\frac{1}{2}$ **Medium · the standard question**

7 PROBLEMS

1. $\log_3 81$

ANSWER 4

2. $\log_5 \frac{1}{125}$

ANSWER -3

3. $\log_8 2$

ANSWER $\frac{1}{3}$

4. Solve $\log_2 x = 6$

ANSWER $x = 64$

5. Solve $\log_x 49 = 2$

ANSWER $x = 7$

6. Domain of $\log_3(x - 4)$

ANSWER $x > 4$

7. $2^{(\log_2 9)}$

ANSWER 9

Hard · stretch and reason

7 PROBLEMS

1. $\log_4 \frac{1}{32}$

ANSWER -2.5

2. $\log_{(0.5)} 8$

ANSWER -3

3. Solve $\log_x 8 = \frac{3}{2}$

ANSWER $x = 4$

4. Domain of $\log_2(x^2 - 9)$

ANSWER $x < -3$ or $x > 3$

5. Domain of $\log_5(6 - x) + \log_5(x - 1)$

ANSWER $1 < x < 6$

6. Solve $3^{\log_3 5 + 1}$

ANSWER 15

7. Explain why $\log_2(-4)$ does not exist

ANSWER No power of 2 is negative

07 Homework

Homework — 5 tasks

Read every logarithm aloud as a question before answering it.

1. Evaluate $\log_2 64$, $\log_3 \frac{1}{27}$, $\log_{16} 4$ and $\lg 0.001$.
2. Solve $\log_5 x = 3$ and $\log_x 64 = 3$.
3. Find the domain of $y = \log_4(2x - 6)$ and of $y = \log_2(x^2 - 4)$.
4. Sketch $y = \log_2 x$ and $y = 2^x$ on one set of axes with the line $y = x$.

5. Explain in three sentences why the argument of a logarithm must be positive.

Transforming logarithmic expressions

Three laws that turn products into sums, and the change of base that lets a calculator finish the job.

LESSON 59–60

2 × 40 MIN

ALGEBRA 10, §3.5

P2 · 2.2

LESSON CLOCK

12'

22'

22'

22'

20'

12'

The three laws	12 min
Expanding and condensing	22 min
Change of base	22 min
The three traps	22 min
Practice	20 min
Homework	12 min

LEARNING OBJECTIVES

- State and apply the three laws of logarithms.
- Expand and condense logarithmic expressions.
- Use the change-of-base formula.
- Recognise and avoid the three standard misapplications.

TEXTBOOK

National	pp. 237–248
Cambridge	Pure Mathematics 2 & 3, pp. 18–25
Workbook	P2 Exercise 2E

01

Explanation

The three laws

$\log_a(xy) = \log_a x + \log_a y$

$$\log_a\left(\frac{x}{y}\right) = \log_a x - \log_a y$$

$$\log_a(x^n) = n \cdot \log_a x$$

WHERE THEY COME FROM

A logarithm is an index, and $a^m \cdot a^n = a^{(m+n)}$. Multiplying the numbers **adds** the indices — so multiplying the arguments adds the logarithms. All three laws are index laws in disguise.

Before calculators, this is exactly how multiplication was done: look up two logarithms, add them, look the answer back up. The slide rule is a physical version of

the first law.

Expanding and condensing

EXPRESSION	EXPANDED
$\log(x^2y^3)$	$2 \log x + 3 \log y$
$\log\left(\frac{x}{y^2}\right)$	$\log x - 2 \log y$
$\log(x\sqrt{y})$	$\log x + \frac{1}{2} \log y$
$\log\left(\frac{x^3}{\sqrt{y}}\right)$	$3 \log x - \frac{1}{2} \log y$

Going the other way — **condensing** — is what solves equations. Coefficients become powers, sums become products, differences become quotients:

$$2 \log x + \log 3 - \log y = \log \frac{3x^2}{y}$$

Change of base

$$\log_a b = \frac{\log_c b}{\log_c a}$$

Any convenient base c may be used — usually 10 or e , because those are the two on a calculator:

$$\log_2 10 = \frac{\lg 10}{\lg 2} = \frac{1}{0.30103} \approx 3.32$$

Two useful special cases follow at once:

$$\log_a b = \frac{1}{\log_b a} \quad \log_{(a^n)} b = \frac{1}{n} \log_a b$$

The three traps

TRAP 1

$\log(x + y)$ is **not** $\log x + \log y$. There is no law for the logarithm of a sum. Test it: $\lg(10 + 10) = \lg 20 \approx 1.3$, but $\lg 10 + \lg 10 = 2$.

TRAP 2

$\frac{\log x}{\log y}$ is **not** $\log\left(\frac{x}{y}\right)$. The quotient law applies to the logarithm **of** a quotient, not to a quotient **of** logarithms.

TRAP 3

$(\log x)^2$ is not $\log(x^2)$. The first is a logarithm squared; the second is $2 \log x$. Watch where the bracket is.

These three account for almost every wrong answer in the topic. Read the expression, decide which law **could** apply, and if none does, leave it alone.

02 Worked examples

EXAMPLE 1 Expand $\log_2 \frac{8x^3}{\sqrt{y}}$.

$$(1) \quad \log_2 8 + \log_2 x^3 - \log_2 \sqrt{y}$$

$$(2) \quad = 3 + 3 \log_2 x - \frac{1}{2} \log_2 y$$

ANSWER

$$3 + 3 \log_2 x - \frac{1}{2} \log_2 y$$

EXAMPLE 2 Condense $3 \log x - 2 \log y + \frac{1}{2} \log z$.

$$\textcircled{1} \quad \log x^3 - \log y^2 + \log \sqrt{z}$$

$$\textcircled{2} \quad = \log \frac{x^3 \sqrt{z}}{y^2}$$

ANSWER

$$\log \frac{x^3 \sqrt{z}}{y^2}$$

EXAMPLE 3 Given $\lg 2 \approx 0.301$, find $\log_2 5$.

$$\textcircled{1} \quad \lg 5 = \lg \frac{10}{2} = 1 - 0.301 = 0.699$$

$$\textcircled{2} \quad \log_2 5 = \frac{\lg 5}{\lg 2} = \frac{0.699}{0.301}$$

$$\textcircled{3} \quad \approx 2.32$$

ANSWER

$$\approx 2.32$$

03 Interactive model

Test every "law" a learner invents on numbers before accepting it.

Is that a law?

$\log(xy)$ equals:

$$\log x \cdot \log y$$

$$\log x + \log y$$

$$\log x - \log y$$

$$(\log x)(\log y)$$

$\log(x + y)$ equals:

$$\log x + \log y$$

$$\log x \cdot \log y$$

none of these

$$\log x - \log y$$

$\log(x^5)$ equals:

$$5 \log x$$

$$(\log x)^5$$

$$\log 5 + \log x$$

$$\frac{\log x}{5}$$

$\log x$ $\log y$ equals:

$$\log\left(\frac{x}{y}\right)$$

$$\log x - \log y$$

$$\log_y x$$

$$\log x \cdot \log y$$

$\log_a b$ in base 10:

$$\frac{\lg a}{\lg b}$$

$$\frac{\lg b}{\lg a}$$

$$\lg b - \lg a$$

$$\lg(ab)$$

$(\log x)^2$ equals:

$$2 \log x$$

$$\log(x^2)$$

neither

$$\log x + \log x$$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Law of logarithms	Logarifm qoidasi	Свойство логарифма
Product law	Ko'paytma qoidasi	Логарифм произведения
Quotient law	Bo'linma qoidasi	Логарифм частного
Power law	Daraja qoidasi	Логарифм степени
Change of base	Asosni almashtirish	Переход к новому основанию
Expand	Yoyish	Разложить
Condense	Bir logarifmga keltirish	Свернуть
Common logarithm	O'nli logarifm	Десятичный логарифм
Misapplication	Noto'g'ri qo'llash	Неверное применение

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

$\log_a x + \log_a y$ equals:

$$\log_a(x + y)$$

$$\log_a(xy)$$

$$\log_a\left(\frac{x}{y}\right)$$

$$2\log_a x$$

$3 \log x$ equals:

$$\log 3x$$

$$\log x^3$$

$$(\log x)^3$$

$$\log 3 + \log x$$

$\log_5 25 - \log_5 5$ equals:

$$1$$

$$5$$

$$20$$

$$\log_5 20$$

Change of base $\log_a b$ is:

$$\frac{\log_c a}{\log_c b}$$

$$\frac{\log_c b}{\log_c a}$$

$$\log_c b \cdot \log_c a$$

$$\log_c(ab)$$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $\log_2 4 + \log_2 8$

ANSWER 5

2. $\log_3 27 - \log_3 3$

ANSWER 2

3. $\log_2 2^5$

ANSWER 5

4. Expand $\log(xy)$

ANSWER $\log x + \log y$

5. Expand $\log(x^3)$

ANSWER $3 \log x$

6. Condense $\log 2 + \log 5$

ANSWER $\log 10 = 1$

7. Condense $2 \log x$

ANSWER $\log x^2$

Medium · the standard question

7 PROBLEMS

1. Expand $\log_2 \frac{8x^3}{\sqrt{y}}$

ANSWER $3 + 3\log_2 x - \frac{1}{2}\log_2 y$

2. Condense $3 \log x - 2 \log y$

ANSWER $\log \frac{x^3}{y^2}$

3. $\lg 4 + \lg 25$

ANSWER 2

4. $\log_6 4 + \log_6 9$

ANSWER 2

5. $\log_2 10$ given $\lg 2 = 0.301$

ANSWER ≈ 3.32

6. $\log_2 5$ given $\lg 2 = 0.301$

ANSWER ≈ 2.32

7. $\log_9 27$

ANSWER 1.5

Hard · stretch and reason

7 PROBLEMS

1. Simplify $\log_2 3 \cdot \log_3 8$

ANSWER 3

2. Simplify $\frac{1}{\log_2 6} + \frac{1}{\log_3 6}$

ANSWER 1

3. Given $\lg 2 = a$ and $\lg 3 = b$, express $\lg 12$

ANSWER $2a + b$

4. Same: express $\lg 1.5$

ANSWER $b - a$

5. Simplify $\log_5 2 \cdot \log_4 25$

ANSWER 1

6. Solve $\log_2 x + \log_2(x - 2) = 3$

ANSWER $x = 4$

7. Show $\log_a b \cdot \log_b a = 1$

ANSWER Change of base twice

07 Homework

Homework — 6 tasks

Test any rule you are unsure of on actual numbers first.

- Expand $\log_3 \frac{9x^4}{y^2}$ and $\lg(x\sqrt{y})$.
- Condense $4 \log x + \frac{1}{2} \log y - 3 \log z$.
- Evaluate $\log_2 6 + \log_2 \frac{8}{3}$ and $\lg 50 + \lg 2$.

4. Given $\lg 2 = 0.301$ and $\lg 3 = 0.477$, find $\lg 6$, $\lg 1.5$ and $\log_3 2$.
5. Show that $\log_a b \cdot \log_b c \cdot \log_c a = 1$.
6. Give a numerical counterexample to each of the three traps.

Logarithmic equations

The domain first, the laws second, the check last — and no step in a different order.

LESSON 61–62

2 × 40 MIN

ALGEBRA 10, §3.6

P2 · 2.4

LESSON CLOCK

12'

22'

22'

22'

20'

12'

The domain first	12 min
Equating arguments	22 min
Exponential form	22 min
Substitution	22 min
Practice	20 min
Homework	12 min

LEARNING OBJECTIVES

- State the domain of a logarithmic equation before solving it.
- Solve $\log_a f(x) = \log_a g(x)$ by equating arguments.
- Solve equations by converting to exponential form.
- Reject roots that fall outside the domain.

TEXTBOOK

National	pp. 249–260
Cambridge	Pure Mathematics 2 & 3, pp. 39–44
Workbook	P2 Exercise 2F

01

Explanation

The domain first

EVERY ARGUMENT MUST BE POSITIVE

Before any algebra, write the conditions. For $\log_2(x - 1) + \log_2(x + 2) = 2$: $x - 1 > 0$ and $x + 2 > 0$, so $x > 1$.

Condensing the two logarithms into $\log_2[(x-1)(x+2)]$ **widens** the domain — the product is also positive when both factors are negative. That is exactly where extraneous roots come from, and the domain line written first is what removes them.

THE CONDENSING STEP IS NOT REVERSIBLE

$\log A + \log B = \log(AB)$ is true only where both A and B are positive. Used left to right in an equation, it is a one-way street.

Equating arguments

$$\log_a f(x) = \log_a g(x) \Rightarrow f(x) = g(x)$$

Valid because the logarithm is one-to-one — the same property that solved exponential equations, used in the other direction.

$$\log_3(2x - 1) = \log_3(x + 4) \Rightarrow 2x - 1 = x + 4 \Rightarrow x = 5$$

Check: $2(5) - 1 = 9 > 0$ and $5 + 4 = 9 > 0$ ✓.

Converting to exponential form

When one side is a number rather than a logarithm, use the definition:

$$\log_a f(x) = c \Rightarrow f(x) = a^c$$

$$\log_2(x - 1) + \log_2(x + 2) = 2 \Rightarrow \log_2[(x-1)(x+2)] = 2 \Rightarrow (x-1)(x+2) = 4$$

So $x^2 + x - 6 = 0$, giving $x = 2$ or $x = -3$. The domain was $x > 1$, so $x = -3$ is rejected and only $x = 2$ survives.

THE SHAPE OF A FULL SOLUTION

1 Domain. **2** Condense to a single logarithm. **3** Convert or equate. **4** Solve. **5** Reject anything outside the domain, and say why.

Substitution

When the same logarithm appears twice, name it:

$$(\lg x)^2 - 3 \lg x + 2 = 0, t = \lg x$$

Then $t^2 - 3t + 2 = 0$ gives $t = 1$ or $t = 2$, so $x = 10$ or $x = 100$. Both are positive, so both stand.

Note that t here has **no** sign restriction — a logarithm may be any real number. That is the opposite of the exponential substitution, and confusing the two is a common error.

02 Worked examples

EXAMPLE 1 Solve $\log_5(3x - 2) = 2$.

① Domain: $3x - 2 > 0 \Rightarrow x > \frac{2}{3}$.

② $3x - 2 = 25$

③ $x = 9$. In the domain ✓

ANSWER

$x = 9$

EXAMPLE 2 Solve $\log_2(x - 1) + \log_2(x + 2) = 2$.

① Domain: $x > 1$.
Both arguments positive.

② $(x - 1)(x + 2) = 4$

③ $x^2 + x - 6 = 0 \Rightarrow x = 2, -3$

④ $x = -3$ fails the domain.

ANSWER

$x = 2$

EXAMPLE 3 Solve $(\lg x)^2 - 3 \lg x + 2 = 0$.

- ① Domain $x > 0$. Let $t = \lg x$.
No sign condition on t .

② $t^2 - 3t + 2 = 0$

③ $t = 1, 2$

④ $x = 10$ or $x = 100$.

ANSWER

$x = 10$ and $x = 100$

03 Interactive model

Write the domain in a box at the top of the board and leave it there.

Domain firstDomain of $\log_2(x - 3) = 1$:

$x > 0$

$x > 3$

$x \geq 3$

$\mathbb{R}$

 $\log_5(3x - 2) = 2$ gives:

$x = 9$

$x = 4$

$x = 27$

$x = 3$

 $\log_2(x-1) + \log_2(x+2) = 2$ gives:

$x = 2, -3$

$x = 2$

$x = -3$

no solution

In $t = \lg x$ the condition on t is:

$\log_3(2x-1) = \log_3(x+4)$ gives:

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Logarithmic equation	Logarifmik tenglama	Логарифмическое уравнение
Domain of definition	Aniqlanish sohasi	Область допустимых значений
Equating arguments	Argumentlarni tenglashtirish	Приравнивание аргументов
Exponential form	Ko'rsatkichli shakl	Показательная форма
Extraneous root	Chet ildiz	Посторонний корень
Condensing	Bir logarifmga keltirish	Свёртывание
Substitution	Almashtirish	Замена
Verification	Tekshirish	Проверка

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The first line of a logarithmic equation is:

the law used

the domain

the answer

the check

$\log_a f = \log_a g$ gives $f = g$ because the logarithm is:

positive

one-to-one

continuous

even

Condensing two logarithms:

never changes the domain

can widen the domain

narrows the domain

is illegal

$\log_2 x = 5$ gives:

$$x = 10$$

$$x = 25$$

$$x = 32$$

$$x = 2.5$$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Solve $\log_2 x = 4$

ANSWER $x = 16$

2. Solve $\log_3 x = 2$

ANSWER $x = 9$

3. Solve $\lg x = 3$

ANSWER $x = 1000$

4. Solve $\log_5(x + 1) = 1$

ANSWER $x = 4$

5. Domain of $\log_2(x - 5)$

ANSWER $x > 5$

6. Solve $\log_2 x = 0$

ANSWER $x = 1$

7. Solve $\log_4 x = \frac{1}{2}$

ANSWER $x = 2$

Medium · the standard question

7 PROBLEMS

1. Solve $\log_5(3x - 2) = 2$

ANSWER $x = 9$

2. Solve $\log_3(2x - 1) = \log_3(x + 4)$

ANSWER $x = 5$

3. Solve $\log_2(x - 1) + \log_2(x + 2) = 2$

ANSWER $x = 2$

4. Solve $\lg x + \lg(x - 3) = 1$

ANSWER $x = 5$

5. Solve $(\lg x)^2 - 3 \lg x + 2 = 0$

ANSWER $x = 10, 100$

6. Solve $\log_2(x + 3) - \log_2(x - 1) = 2$

ANSWER $x = \frac{7}{3}$

7. Solve $\log_3(x^2) = 4$

ANSWER $x = \pm 9$

Hard · stretch and reason

7 PROBLEMS

1. Solve $\log_2(x - 2) + \log_2(x - 3) = 1$

ANSWER $x = 4$

2. Solve $(\log_2 x)^2 - \log_2 x - 6 = 0$

ANSWER $x = 8$ or $x = \frac{1}{4}$

3. Solve $\lg(x + 6) - \frac{1}{2} \lg(2x - 3) = 2 - \lg 25$

ANSWER $x = 6$

4. Solve $\log_x 4 + \log_x 16 = 3$

ANSWER $x = 4$

5. Solve $\log_3 x + \log_x 3 = 2.5$

ANSWER $x = 9$ or $x = \sqrt{3}$

6. Solve $\lg^2 x - \lg x^3 + 2 = 0$

ANSWER $x = 10, 100$

7. Explain why $\lg x + \lg(x - 3) = 1$ has only one root

ANSWER $x = -2$ fails the domain

07 Homework

Homework — 6 tasks

Open every solution with the domain, and close it with the rejection.

1. Solve $\log_4(2x + 3) = 2$.

2. Solve $\log_2(x + 5) = \log_2(3x - 1)$.

3. Solve $\lg x + \lg(x - 15) = 2$.

4. Solve $(\log_3 x)^2 - 4 \log_3 x + 3 = 0$.

5. Solve $\log_5(x + 4) - \log_5(x - 2) = 1$.
6. Explain in three sentences why condensing two logarithms can create a root the original never had.

Systems of exponential and logarithmic equations

Two equations, one substitution — and the domain that has to survive both.

LESSON 63–64

2 × 40 MIN

ALGEBRA 10, §3.7

EXTENSION

LESSON CLOCK



What is new	10 min
Exponential systems	24 min
Logarithmic systems	24 min
Mixed systems	22 min
Practice	20 min
Homework	10 min

LEARNING OBJECTIVES

- Solve a system by substituting $u = a^x$ and $v = a^y$.
- Solve a system by substituting $u = \log x$ and $v = \log y$.
- Track the domain conditions of both equations.
- Check every candidate pair in the original system.

TEXTBOOK

National	pp. 261–272
Cambridge	Pure Mathematics 2 & 3, pp. 39–44
Workbook	P2 Exercise 2F

01 Explanation

What is new, and what is not

Nothing about the algebra is new: the systems are solved by substitution and elimination exactly as in Quarter II. What is new is the bookkeeping — two domains, two substitutions and two sign conditions to keep straight.

TWO SUBSTITUTIONS, TWO DIFFERENT CONDITIONS

$u = a^x$ carries $u > 0$. $u = \log x$ carries **no** condition on u , but the original x must satisfy $x > 0$. Write the right one down.

Exponential systems

$$2^x + 3^y = 17 \quad 2^{x+2} - 3^{y+1} = 5$$

Put $u = 2^x > 0$ and $v = 3^y > 0$. The second equation is $4u - 3v = 5$, so with $u + v = 17$:

$$4u - 3(17 - u) = 5 \Rightarrow 7u = 56 \Rightarrow u = 8, v = 9$$

Both positive, so $2^x = 8$ gives $x = 3$ and $3^y = 9$ gives $y = 2$.

EXPAND THE INDICES FIRST

$2^{x+2} = 4 \cdot 2^x$, not $2^x + 4$. Every index law is applied before the substitution, not after.

Logarithmic systems

$$\lg x + \lg y = 3 \lg x - \lg y = 1$$

Domain $x > 0, y > 0$. Adding gives $2 \lg x = 4$, so $\lg x = 2$ and $x = 100$; then $\lg y = 1$ and $y = 10$. Both positive ✓.

An equivalent route condenses first: $\lg(xy) = 3$ gives $xy = 1000$, and $\lg \frac{x}{y} = 1$ gives

$\frac{x}{y} = 10$ — the sum-and-quotient system of Quarter II.

Mixed systems

When one equation is exponential and the other logarithmic, solve the simpler one for a variable and substitute:

$$\log_2 x + y = 5 \quad 2^y = x$$

The second gives $y = \log_2 x$, so the first is $2 \log_2 x = 5$ and $\log_2 x = 2.5$, giving $x = 2^{2.5} = 4\sqrt{2} \approx 5.66$ and $y = 2.5$.

LOOK FOR THE LINK

The two equations almost always share a structure — the same base, or one is the other rearranged. Finding that link is the step; everything after it is Grade 9 algebra.

02 Worked examples

EXAMPLE 1 Solve $2^x + 3^y = 17$, $2^{x+2} - 3^{y+1} = 5$.

① $u = 2^x > 0, v = 3^y > 0$
 $u + v = 17, 4u - 3v = 5$

② $4u - 51 + 3u = 5 \Rightarrow u = 8$

③ $v = 9$

④ $x = 3, y = 2$

ANSWER

$(3, 2)$

EXAMPLE 2 Solve $\lg x + \lg y = 3$, $\lg x - \lg y = 1$.

① Domain $x, y > 0$.

② Add: $2 \lg x = 4 \Rightarrow \lg x = 2$.

③ $x = 100$

④ $\lg y = 1 \Rightarrow y = 10$

ANSWER

$(100, 10)$

EXAMPLE 3 Solve $x + y = 10$, $\lg x + \lg y = 1$.

① Domain $x, y > 0$.

② $\lg(xy) = 1 \Rightarrow xy = 10$

③ $t^2 - 10t + 10 = 0$

④ $t = 5 \pm \sqrt{15}$
Both positive ✓

ANSWER

$(5 + \sqrt{15}, 5 - \sqrt{15})$ and the reverse

03 Interactive model

Write both domains on the board before either substitution.

Which substitution?

For $2^x + 3^y = 17$ put:

$$u = x, v = y$$

$$u = 2^x, v = 3^y$$

$$u = \lg x$$

nothing

The condition on that u is:

$$u \geq 0$$

$$u > 0$$

none

$$u \neq 1$$

2^{x+2} equals:

$$2^x + 4$$

$$4 \cdot 2^x$$

$$2^x \cdot 2$$

$$(2^x)^2$$

For $\lg x + \lg y = 3$ the condition is:

$$x, y > 0$$

$$x, y \geq 0$$

$$x + y > 0$$

none

$\lg x + \lg y = 3$ means:

$$x + y = 1000$$

$$xy = 1000$$

$$xy = 3$$

$$x + y = 3$$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
System of equations	Tenglamalar sistemasi	Система уравнений
Auxiliary variables	Yordamchi o'zgaruvchilar	Вспомогательные переменные
Combined domain	Umumiy aniqlanish sohasi	Общая область определения
Back-substitution	Teskari almashtirish	Обратная подстановка
Symmetric system	Simmetrik sistema	Симметричная система
Elimination	Yo'qotish	Исключение
Rejected pair	Rad etilgan juftlik	Отброшенная пара
Verification	Tekshirish	Проверка

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

$u = a^x$ carries:

$u = \lg x$ carries:

$3^{(y+1)}$ equals:

$\lg x - \lg y = 1$ means:

$$x - y = 10$$

$$\frac{x}{y} = 10$$

$$\frac{x}{y} = 1$$

$$x - y = 1$$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Solve $2^x = 8, y = x + 1$

ANSWER $(3, 4)$

2. Solve $\lg x = 2, y = 2x$

ANSWER $(100, 200)$

3. Write 2^{x+3} in terms of 2^x

ANSWER $8 \cdot 2^x$

4. Write 3^{y-1} in terms of 3^y

ANSWER $\frac{3^y}{3}$

5. $\lg x + \lg y$ condensed

ANSWER $\lg(xy)$

6. $\lg x - \lg y$ condensed

ANSWER $\lg \frac{x}{y}$

7. Condition for $u = 5^x$

ANSWER $u > 0$

Medium · the standard question

7 PROBLEMS

1. Solve $2^x + 3^y = 17, 2^{x+2} - 3^{y+1} = 5$

ANSWER $(3, 2)$

2. Solve $\lg x + \lg y = 3, \lg x - \lg y = 1$

ANSWER $(100, 10)$

3. Solve $x + y = 10, \lg x + \lg y = 1$

ANSWER $(5 \pm \sqrt{15})$ pairs

4. Solve $2^x \cdot 3^y = 12, 2^x + 3^y = 7$

ANSWER $(2, 1)$ or $(\log_2 3, \log_3 4)$

5. Solve $\log_2 x + y = 5, 2^y = x$

ANSWER $x = 4\sqrt{2}, y = 2.5$

6. Solve $3^x = 9^y, x + y = 3$

ANSWER $(2, 1)$

7. Solve $\lg x = 2 \lg y, x + y = 12$

ANSWER $(9, 3)$

Hard · stretch and reason

7 PROBLEMS

1. Solve $4^x + 4^y = 20, x + y = 3$

ANSWER (1, 2) and (2, 1)

2. Solve $\lg(x + y) = 1, \lg x - \lg y = \lg 4$

ANSWER (8, 2)

3. Solve $2^x + 2^y = 6, 2^{x+y} = 8$

ANSWER (1, 2) and (2, 1)

4. Solve $\log_2 x + \log_2 y = 5, x - y = 6$

ANSWER (8, 4)

5. Solve $x^y = 8, \log_2 x \cdot y = 3$

ANSWER Any x, y with $y \log_2 x = 3$

6. Solve $9^x + 9^y = 90, x + y = 2$

ANSWER (0, 2) and (2, 0)

7. Explain why a pair with $u = -2$ is rejected

ANSWER a^x is never negative

07 Homework

Homework — 5 tasks

Both domains on the first line; both ordered pairs at the end where the system is symmetric.

1. Solve $3^x + 2^y = 17, 3^{x+1} - 2^{y+2} = 7$.
2. Solve $\lg x + \lg y = 4, \lg x - \lg y = 2$.
3. Solve $x + y = 20, \lg x + \lg y = 2$.
4. Solve $2^x = 4^y, x + y = 6$.
5. Explain in three sentences why $u = a^x$ needs a sign condition but $u = \lg x$ does not.

Logarithmic inequalities

Domain, then direction, then solve — and the intersection at the end that catches everything.

LESSON 65–66

2 × 40 MIN

ALGEBRA 10, §3.8

P2 · 2.4

LESSON CLOCK



Three steps, in order 12 min

Base above 1 22 min

Base below 1 22 min

Substitution 22 min

Practice 20 min

Homework 12 min

LEARNING OBJECTIVES

- State the domain of a logarithmic inequality before solving.
- Use monotonicity to compare arguments, reversing when the base is below 1.
- Intersect the solution with the domain.
- Solve inequalities that become quadratic after a substitution.

TEXTBOOK

National pp. 273–284

Cambridge Pure Mathematics
2 & 3, pp. 39–44

Workbook P2 Exercise 2G

01 Explanation

Three steps, in order

NEVER CHANGE THE ORDER

1 Write the domain — every argument strictly positive. **2** Compare the arguments, reversing the sign if the base is under 1. **3 Intersect** the result with the domain.

Step 3 is not optional and is not a check: it is where a large part of the answer is decided. An inequality solved correctly and not intersected is usually wrong.

Base above 1

$$a > 1: \log_a f > \log_a g \Leftrightarrow f > g > 0$$

Example. $\log_2(x - 1) < 3$.

1. Domain: $x > 1$.

2. $\log_2(x - 1) < \log_2 8$, base > 1 , so $x - 1 < 8$ and $x < 9$.

3. Intersect: $1 < x < 9$.

Without step 3 the answer would be $x < 9$, which includes $x = 0$ — where the expression does not exist.

Base below 1

$$0 < a < 1: \log_a f > \log_a g \Leftrightarrow 0 < f < g$$

Example. $\log_{(1/3)}(x + 2) > -1$.

1. Domain: $x > -2$.
2. $-1 = \log_{(1/3)} 3$. Base < 1 , so reverse: $x + 2 < 3$ and $x < 1$.
3. Intersect: $-2 < x < 1$.

TWO THINGS REVERSE, NOT ONE

A base under 1 reverses the comparison of the arguments — and if the inequality is later multiplied by a negative or divided, that reverses again. Do one reversal at a time, and say out loud which one you are doing.

Substitution

With $t = \log_a x$ (no sign condition) the inequality becomes an ordinary quadratic one:

$$(\lg x)^2 - 3 \lg x + 2 < 0, t = \lg x \Rightarrow 1 < t < 2$$

$$1 < \lg x < 2 \Rightarrow 10 < x < 100$$

The base 10 exceeds 1, so the order is preserved. With a base under 1 the whole double inequality would flip.

THE THREE SOURCES OF ERROR, IN ORDER OF FREQUENCY

forgetting the intersection · forgetting the reversal · forgetting that t has no sign condition. Address them in that order.

02 Worked examples

EXAMPLE 1 Solve $\log_2(x - 1) < 3$.

① Domain $x > 1$.

② $x - 1 < 8$
Base > 1 .

③ $x < 9$

④ Intersect.

ANSWER

$$1 < x < 9$$

EXAMPLE 2 Solve $\log_{(1/3)}(x + 2) > -1$.

① Domain $x > -2$.

② $-1 = \log_{(1/3)} 3$

③ Base < 1 : reverse. $x + 2 < 3$.

④ $x < 1$; intersect.

ANSWER

$$-2 < x < 1$$

EXAMPLE 3 Solve $\lg x + \lg(x - 3) < 1$.

- ① Domain $x > 3$.
Both arguments.
- ② $\lg[x(x - 3)] < \lg 10$
- ③ $x^2 - 3x - 10 < 0 \Rightarrow -2 < x < 5$
- ④ Intersect with $x > 3$.

ANSWER

$$3 < x < 5$$

03 Interactive model

Draw two number lines — the domain and the solution — and shade the overlap.

Domain and solution

$$x > 2$$

inequality $x > 2$ boundary **2 (open)** $x = 0$ works? **no** $x = 6$ works? **yes**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Logarithmic inequality	Logarifmik tengsizlik	Логарифмическое неравенство
Domain of definition	Aniqlanish sohasi	Область допустимых значений
Monotonicity	Monotonlik	Монотонность
Reversing the sign	Ishorani almashtirish	Смена знака
Intersection with the domain	Soha bilan kesishma	Пересечение с ОДЗ
Substitution	Almashtirish	Замена
Solution set	Yechimlar to'plami	Множество решений
Interval notation	Oraliq belgilanishi	Интервальная запись

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The first step is:

compare arguments

write the domain

substitute

reverse

$\log_2(x - 1) < 3$ gives:

$x < 9$

$1 < x < 9$

$x > 1$

$x < 8$

A base under 1:

keeps the sign

reverses the sign

makes it undefined

doubles it

In $t = \lg x$ the condition on t is:

$$t > 0$$

none

$$t \geq 0$$

$$t \neq 0$$

$\lg x < 2$ gives:

$$x < 100$$

$$0 < x < 100$$

$$x > 100$$

$$x < 2$$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Solve $\lg x > 1$

ANSWER $x > 10$

2. Solve $\lg x < 2$

ANSWER $0 < x < 100$

3. Solve $\log_2 x > 3$

ANSWER $x > 8$

4. Solve $\log_2 x < 0$

ANSWER $0 < x < 1$

5. Domain of $\log_3(x - 5)$

ANSWER $x > 5$

6. Solve $\log_5 x \geq 1$

ANSWER $x \geq 5$

7. Solve $\log_{(1/2)} x > 0$

ANSWER $0 < x < 1$

Medium · the standard question

7 PROBLEMS

1. Solve $\log_2(x - 1) < 3$

ANSWER $1 < x < 9$

2. Solve $\log_3(2x + 1) > 2$

ANSWER $x > 4$

3. Solve $\log_{1/3}(x + 2) > -1$

ANSWER $-2 < x < 1$

4. Solve $\lg x + \lg(x - 3) < 1$

ANSWER $3 < x < 5$

5. Solve $\log_2(x + 3) \leq 4$

ANSWER $-3 < x \leq 13$

6. Solve $\log_{0.5}(x - 1) \geq -2$

ANSWER $1 < x \leq 5$

7. Solve $\log_4(x^2) < 1$

ANSWER $-2 < x < 2, x \neq 0$

Hard · stretch and reason

7 PROBLEMS

1. Solve $(\lg x)^2 - 3 \lg x + 2 < 0$

ANSWER $10 < x < 100$

2. Solve $(\log_2 x)^2 - \log_2 x - 6 \geq 0$

ANSWER $0 < x \leq \frac{1}{4}$ or $x \geq 8$

3. Solve $\log_2(x - 1) + \log_2(x + 1) < 3$

ANSWER $1 < x < 3$

4. Solve $\log_{-}(1/2)(x^2 - 3x) \geq -2$

ANSWER $-1 \leq x < 0$ or $3 < x \leq 4$

5. Solve $\log_3(x + 1) > \log_3(2x - 3)$

ANSWER $1.5 < x < 4$

6. Solve $\lg(x - 2) < \lg(4 - x)$

ANSWER $2 < x < 3$

7. Explain why $\log_2 x < 3$ is not simply $x < 8$

ANSWER The domain $x > 0$ must be intersected

07 Homework

Homework — 6 tasks

Draw the domain and the solution on the same number line every time.

1. Solve $\log_3(x - 2) < 2$.

2. Solve $\log_{-}(1/4)(x + 1) > -1$.

3. Solve $\lg x + \lg(x - 21) < 2$.

4. Solve $(\log_3 x)^2 - 4 \log_3 x + 3 \leq 0$.

5. Solve $\log_2(3x - 1) \geq \log_2(x + 5)$.

6. Explain in three sentences why the intersection with the domain is part of the method and not a check.

Applications of exponential and logarithmic functions

Compound interest, half-lives, earthquakes and sound — the places where these two functions are not optional.

LESSON 67–68

2 × 40 MIN

ALGEBRA 10, §3.9

P2 · 2.1, 2.5

LESSON CLOCK



Growth, and solving for the time	16 min
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Decay and half-life	22 min
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Logarithmic scales	22 min
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Linearising	20 min
-------------	--------

Practice	18 min
----------	--------

Homework	10 min
----------	--------

LEARNING OBJECTIVES

- Model compound growth and decay, and solve for the time.
- Use a logarithm to bring an unknown out of an index.
- Interpret a logarithmic scale (pH, decibels, Richter).
- Linearise an exponential model by taking logarithms.

TEXTBOOK

National	pp. 285–296
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Cambridge	Pure Mathematics 2 & 3, pp. 26–32, 47–52
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Workbook	P2 Exercise 2H
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01 Explanation

Growth, and solving for the time

$$A = P(1 + r)^n$$

P the starting amount, r the rate per period as a decimal, n the number of periods.

Finding A is arithmetic; finding n needs a logarithm:

$$n = \frac{\lg(A/P)}{\lg(1 + r)}$$

Example. 5 million so'm at 14% a year. When does it double?

$$1.14^n = 2 \Rightarrow n = \frac{\lg 2}{\lg 1.14} = \frac{0.3010}{0.0569} \approx 5.29 \text{ years}$$

WHY THE LOGARITHM IS NEEDED

The unknown is in the index. No amount of rearranging brings it down — only taking a logarithm of both sides does. That is the practical reason this whole chapter exists.

Decay and half-life

$$N = N_0 \cdot (\frac{1}{2})^{(t/h)}$$
where h is the half-life

Solving for t again needs a logarithm:

$$t = h \cdot \frac{\lg(N/N_0)}{\lg 0.5}$$

SUBSTANCE	HALF-LIFE	USE
carbon-14	5730 years	dating organic remains
iodine-131	8 days	medical imaging
caffeine in the body	about 5 hours	why coffee keeps you awake

Logarithmic scales

When a quantity ranges over many powers of ten, its logarithm is the readable number:

SCALE	DEFINITION	ONE UNIT MEANS
Richter	$M = \lg(\frac{A}{A_0})$	10 times the amplitude
decibel	$L = 10 \lg(\frac{I}{I_0})$	10 dB is 10 times the intensity
pH	$pH = -\lg[H^+]$	10 times the concentration

An earthquake of magnitude 7 is not slightly worse than one of 6 — it is 10 times the amplitude and about 32 times the energy.

Linearising

Given data believed to follow $y = k \cdot a^x$, take logarithms of both sides:

$$\lg y = \lg k + x \cdot \lg a$$

Plotting $\lg y$ against x gives a **straight line** of gradient $\lg a$ and intercept $\lg k$. So a ruler on the transformed plot finds both parameters — and a straight plot is itself the evidence that the model is exponential.

For a power law $y = k \cdot x^n$, plot $\lg y$ against $\lg x$ instead: the gradient is then n .

02 Worked examples

EXAMPLE 1 5 million so'm is invested at 14% a year. How long until it doubles?

$$1.14^n = 2$$

$$n \lg 1.14 = \lg 2$$

$$n = \frac{0.3010}{0.0569}$$

$$\approx 5.3 \text{ years.}$$

ANSWER

≈ 5.3 years

EXAMPLE 2 A sample has half-life 8 days. What percentage remains after 20 days?

$$① \quad \left(\frac{1}{2}\right)^{(20/8)} = \left(\frac{1}{2}\right)^{2.5}$$

$$② \quad = \frac{1}{4\sqrt{2}} \approx 0.1768$$

$$③ \quad \approx 17.7\%$$

ANSWER

$\approx 17.7\%$

EXAMPLE 3 One earthquake has magnitude 6.4, another 7.9. Compare their amplitudes.

$$① \quad M = \lg \frac{A}{A_0}$$

$$② \quad 7.9 - 6.4 = 1.5$$

$$③ \quad 10^{1.5} \approx 31.6$$

ANSWER

About 32 times greater

03 Interactive model

Ask for the doubling time of any growth rate mentioned in the news.

Bringing the unknown down

To solve $1.14^n = 2$ you should:

divide by 1.14

take logarithms

square it

guess

5 million at 14%: doubling time is about:

3 y

5.3 y

7 y

14 y

Half-life 8 days: after 24 days there remains:

$\frac{1}{3}$

$\frac{1}{8}$

$\frac{1}{16}$

$\frac{1}{24}$

One Richter unit is a factor of:

2

10

100

32

To find a in $y = k a^x$ from data, plot:

y vs x

$\lg y$ vs x

$\lg y$ vs $\lg x$

y vs $\lg x$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Compound interest	Murakkab foiz	Сложные проценты
Growth factor	O'sish koeffitsienti	Коэффициент роста
Half-life	Yarim yemirilish davri	Период полураспада
Doubling time	Ikkilanish vaqti	Время удвоения
Logarithmic scale	Logarifmik shkala	Логарифмическая шкала
Richter scale	Rixter shkalasi	Шкала Рихтера
Decibel	Detsibel	Децибел
Linearisation	Chiziqlashtirish	Линеаризация
Model parameter	Model parametri	Параметр модели

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Compound interest is modelled by:

$$P + rn$$

$$P(1 + r)^n$$

$$P \cdot rn$$

$$P + r^n$$

To bring an unknown out of an index:

divide

take a logarithm

square

differentiate

Two half-lives leave:

$$\frac{1}{2}$$

$$\frac{1}{4}$$

$$\frac{1}{3}$$

nothing

For a power law plot:

$\lg y$ vs x

$\lg y$ vs $\lg x$

y vs x

y vs $\lg x$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. *1000* at 10% for 3 years

ANSWER *1331*

2. *2000* at 5% for 2 years

ANSWER *2205*

3. Half-life 4 days; fraction after 8 days

ANSWER $\frac{1}{4}$

4. Half-life 4 days; fraction after 12 days

ANSWER $\frac{1}{8}$

5. One Richter unit is a factor of

ANSWER *10*

6. *10* dB is a factor of

ANSWER *10 in intensity*

7. pH 4 versus pH 6: how many times more acidic?

ANSWER *100*

Medium · the standard question

7 PROBLEMS

1. 5 million at 14%: doubling time

ANSWER ≈ 5.3 years

2. 20 000 at 8%: when does it reach 50 000?

ANSWER ≈ 11.9 years

3. Half-life 8 days: percentage after 20 days

ANSWER $\approx 17.7\%$

4. Magnitudes 6.4 and 7.9: amplitude ratio

ANSWER ≈ 32

5. A population trebles in 12 years. Annual rate?

ANSWER $\approx 9.6\%$

6. A car falls 18% a year. When is it worth half?

ANSWER ≈ 3.5 years

7. Sound at 85 dB versus 65 dB: intensity ratio

ANSWER 100

Hard · stretch and reason

7 PROBLEMS

1. Carbon-14 half-life 5730 y; 30% remains. Age?

ANSWER ≈ 9950 years

2. P at r compounded monthly: the annual growth factor

ANSWER $(1 + \frac{r}{12})^{12}$

3. 10% compounded monthly versus annually on 1000 for 1 year

ANSWER 1104.71 versus 1100

4. Data fits $y = k a^x$ with $\lg y$ line of gradient 0.3 and intercept 1. Find k, a

ANSWER $k = 10, a \approx 2$

5. Data fits $y = k x^n$ with $\lg y$ vs $\lg x$ gradient 1.5, intercept 0.6. Find k, n

ANSWER $n = 1.5, k \approx 3.98$

6. Show the doubling time at rate r is about $\frac{70}{100r}$

ANSWER The "rule of 70" from $\frac{\ln 2}{r}$

7. A drug halves every 5 h. What fraction remains after 24 h?

ANSWER $\approx 3.6\%$

07 Homework

Homework — 6 tasks

Every answer needs its units and a sentence in the words of the question.

- 8 million so'm at 12% a year. Find the value after 6 years and the doubling time.
- A sample has half-life 6 hours. Find the percentage remaining after 20 hours.
- Two earthquakes measure 5.8 and 7.3. Compare their amplitudes.
- A population of 12 000 grows 3% a year. When does it pass 20 000?

5. Data believed to follow $y = k a^x$ gives a $\lg y$ line of gradient 0.48 and intercept 0.7 . Find k and a .
6. Explain in three sentences why a logarithmic scale is used for earthquakes.

Control work 6, and work on the mistakes

The logarithmic block in one paper — where the domain line is worth more than the arithmetic.

LESSON 69–70

2 × 40 MIN

ALGEBRA 10, NAZORAT ISHI 6

P2 · CHAPTER 2 REVIEW

LESSON CLOCK

3

40'

10'

25'

12'

Instructions	3 min
The paper	40 min
Self-mark	10 min
Rewrite	25 min
The domain drill	12 min

LEARNING OBJECTIVES

- Apply the logarithmic methods under time.
- State a domain and intersect with it without prompting.
- Classify each lost mark as careless, method or knowledge.
- Rewrite every wrong solution correctly.

TEXTBOOK

National	pp. 297–300
Cambridge	Pure Mathematics 2 & 3, pp. 53–54
Workbook	Control paper F

01

Explanation

The paper — 25 marks, 40 minutes

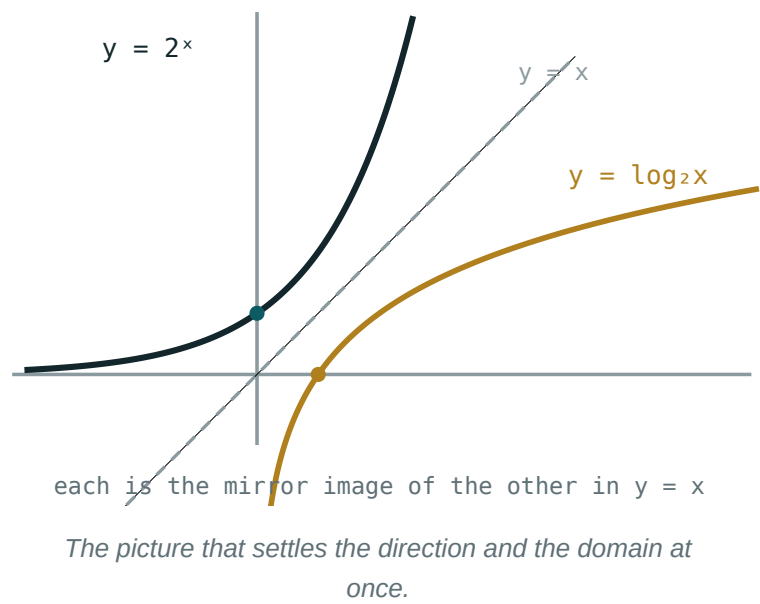
Q	TASK	MARKS	FROM
1	Evaluate $\log_2 32$, $\log_9 3$, $\log_5 \frac{1}{25}$ and $\lg 0.01$	4	L57–58
2	Expand $\log_2 \frac{4x^3}{\sqrt{y}}$ and condense $3 \lg x - 2 \lg y$	4	L59–60
3	Solve $\lg x + \lg(x - 3) = 1$	4	L61–62
4	Solve $(\log_2 x)^2 - 5 \log_2 x + 6 = 0$	4	L61–62
5	Solve $\log_3(x - 1) < 2$ and $\log_{-}(1/2)(x + 3) \geq -2$	5	L65–66
6	6 million at 11% a year: find the doubling time	4	L67–68

FOUR MARKS ARE FOR THE DOMAIN

Q3 and Q5 each give one mark for the domain line and one for the intersection or rejection. A correct final interval with no domain shown scores half.

The four errors

ERROR	LOOKS LIKE	KIND
no intersection	$\log_3(x-1) < 2 \Rightarrow x < 10$, keeping $x = 0$	method
invented law	$\log(x + y) = \log x + \log y$	knowledge
sign not reversed	$\log_{(1/2)}$ treated as base > 1	knowledge
root not rejected	$x = -2$ kept in $\lg x + \lg(x-3) = 1$	method



The domain drill

Twelve expressions on the board. For each, the class calls out only **the domain**, in three seconds:

EXPRESSION	DOMAIN
$\log_2(x - 4)$	$x > 4$
$\lg(5 - x)$	$x < 5$
$\log_3(x^2)$	$x \neq 0$
$\log_5(x^2 - 4)$	$x < -2 \text{ or } x > 2$
$\log_x 9$	$x > 0, x \neq 1$
$\lg x + \lg(x - 3)$	$x > 3$

ROWS 5 AND 6 ARE THE ONES THEY MISS

A variable **base** has its own conditions. And two logarithms added give the **intersection** of the two domains, not the domain of the product.

02 Worked examples

EXAMPLE 1 Model answer, Q3: solve $\lg x + \lg(x - 3) = 1$.

① Domain: $x > 0$ and $x > 3$, so $x > 3$.

One mark.

② $\lg[x(x - 3)] = \lg 10$

③ $x^2 - 3x - 10 = 0 \Rightarrow x = 5, -2$

④ $x = -2$ fails the domain.

One mark.

ANSWER

$x = 5$

EXAMPLE 2 Model answer, Q5b: solve $\log_{(1/2)}(x + 3) \geq -2$.

① Domain $x > -3$.

② $-2 = \log_{(1/2)} 4$

③ Base < 1 : reverse. $x + 3 \leq 4$.

④ $x \leq 1$; intersect.

ANSWER

$$-3 < x \leq 1$$

EXAMPLE 3 Model answer, Q6: 6 million at 11% a year.

① $1.11^n = 2$

② $n = \frac{\lg 2}{\lg 1.11}$

③ $= \frac{0.3010}{0.0453}$

④ ≈ 6.64 years.

ANSWER

$$\approx 6.6 \text{ years}$$

03 Interactive model

Run the domain drill before the rewrite, and again at the end.

Domain, law, direction $\log_2 32$: $\log_9 3$:Domain of $\lg x + \lg(x - 3)$:

$$\lg x + \lg(x - 3) = 1:$$

$$x = 5, -2$$

$$x = 5$$

$$x = -2$$

none

$$(\log_2 x)^2 - 5\log_2 x + 6 = 0:$$

$$x = 2, 3$$

$$x = 4, 8$$

$$x = 6$$

$$x = 8$$

$$\log_3(x - 1) < 2:$$

$$x < 10$$

$$1 < x < 10$$

$$x > 1$$

$$x < 9$$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Control work	Nazorat ishi	Контрольная работа
Domain of definition	Aniqlanish sohasi	ОДЗ
Law of logarithms	Logarifm qoidasi	Свойство логарифма
Intersection	Kesishma	Пересечение
Careless error	E'tiborsizlik xatosi	Ошибка по невнимательности
Method error	Usul xatosi	Ошибка в методе
Knowledge gap	Bilim bo'shlig'i	Пробел в знаниях
Correction	Tuzatish	Исправление

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The domain line is worth:

nothing

a mark on its own

the question

half

$\log(x + y)$ equals:

$\log x + \log y$

$\log x \cdot \log y$

none of these

$\log x - \log y$

A base under 1 in an inequality:

changes nothing

reverses the sign

makes it undefined

doubles it

Adding two logarithms gives a domain that is:

the union

the intersection

the product

wider

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $\log_2 32$

ANSWER 5

2. $\log_9 3$

ANSWER 0.5

3. $\log_5 \frac{1}{25}$

ANSWER -2

4. $\lg 0.01$

ANSWER -2

5. Domain of $\lg(5 - x)$

ANSWER $x < 5$

6. Condense $3 \lg x - 2 \lg y$

ANSWER $\lg \frac{x^3}{y^2}$

7. Solve $\lg x = 2$

ANSWER $x = 100$

Medium · the standard question

7 PROBLEMS

1. Expand $\log_2 \frac{4x^3}{\sqrt{y}}$

ANSWER $2 + 3\log_2 x - \frac{1}{2}\log_2 y$

2. Solve $\lg x + \lg(x - 3) = 1$

ANSWER $x = 5$

3. Solve $(\log_2 x)^2 - 5\log_2 x + 6 = 0$

ANSWER $x = 4, 8$

4. Solve $\log_3(x - 1) < 2$

ANSWER $1 < x < 10$

5. Solve $\log_{1/2}(x + 3) \geq -2$

ANSWER $-3 < x \leq 1$

6. 6 million at 11%: doubling time

ANSWER ≈ 6.6 years

7. Domain of $\log_5(x^2 - 4)$

ANSWER $x < -2$ or $x > 2$

Hard · stretch and reason

7 PROBLEMS

1. Solve $\log_2(x - 1) + \log_2(x + 1) = 3$

ANSWER $x = 3$

2. Solve $\lg^2 x - \lg x^4 + 3 = 0$

ANSWER $x = 10, 1000$

3. Solve $\log_{1/3}(x^2 - 3x) \geq -2$

ANSWER $-1 \leq x < 0$ or $3 < x \leq 4$

4. Solve $\log_x 16 = 4$

ANSWER $x = 2$

5. Solve $\log_2 x + \log_x 2 = 2.5$

ANSWER $x = 4$ or $x = \sqrt{2}$

6. A town of 30 000 grows 4% a year: when does it double?

ANSWER ≈ 17.7 years

7. Explain why $\log_3(x-1) < 2$ is not $x < 10$

ANSWER The domain $x > 1$ must be intersected

07 Homework

Homework — 4 tasks

Task 1 is the rewrite. Domain first on every question.

1. Rewrite in full every question that lost a mark, with the domain first.
2. Five problems from the section your knowledge column was heaviest in.
3. Solve $\lg x + \lg(x - 15) = 2$ and $\log_2(x + 1) \leq 3$.
4. 9 million so'm at 9% a year. Find the value after 5 years and the doubling time.

Trigonometric functions, their properties and graphs

The functions that repeat — and the periodic processes they describe.

LESSON 71–72

2 × 40 MIN

ALGEBRA 10, §4.1

P1 · 5.2

LESSON CLOCK

14'

24'

22'

20'

16'

8'

The unit-circle definition	14 min
The three graphs	24 min
Properties	22 min
Periodic processes	20 min
Practice	16 min
Homework	8 min

LEARNING OBJECTIVES

- Define sine, cosine and tangent on the unit circle for any angle.
- Sketch $y = \sin x$, $y = \cos x$ and $y = \tan x$.
- State the period, range, zeros and symmetry of each.
- Model a periodic process with a trigonometric function.

TEXTBOOK

National	pp. 301–312
Cambridge	Pure Mathematics 1, pp. 96–104
Workbook	P1 Exercise 5B

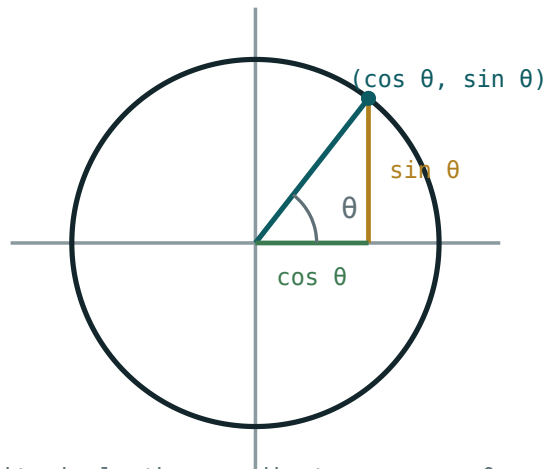
01

Explanation

The unit-circle definition

Take the point P on the unit circle at angle θ from the positive x -axis, measured anticlockwise. Then by definition

$$\cos \theta = x\text{-coordinate of } P \quad \sin \theta = y\text{-coordinate of } P$$



on the unit circle the coordinates are $\cos \theta$ and $\sin \theta$

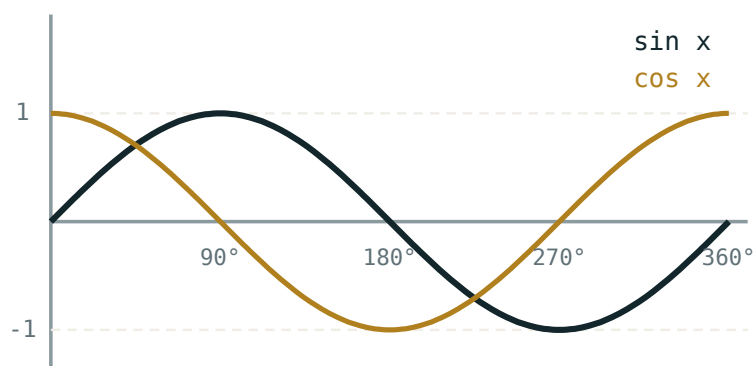
The coordinates of the point are the cosine and the sine. Nothing else is needed.

WHY THIS REPLACES THE RIGHT TRIANGLE

The triangle definition works only for $0^\circ < \theta < 90^\circ$. The circle definition works for every angle — negative, obtuse, reflex, or beyond a full turn — and it makes the signs in each quadrant obvious from the coordinates.

Because P returns to the same place after a full turn, both functions repeat with period 360° . The tangent is $\tan \theta = \frac{\sin \theta}{\cos \theta}$, undefined where $\cos \theta = 0$.

The three graphs



period 360° , range -1 to 1

One period of each. The cosine is the sine shifted left by 90° .

	<i>SIN X</i>	<i>COS X</i>	<i>TAN X</i>
period	360°	360°	180°
domain	$\mathbb{R}$	$\mathbb{R}$	$x \neq 90^\circ + 180^\circ k$
range	$[-1, 1]$	$[-1, 1]$	$\mathbb{R}$
zeros	$180^\circ k$	$90^\circ + 180^\circ k$	$180^\circ k$
parity	odd	even	odd
maximum	1 at 90°	1 at 0°	none

THE TANGENT HAS ASYMPTOTES

At $90^\circ, 270^\circ, \dots$ the cosine is zero and the tangent is undefined. Its graph is a chain of identical branches, each rising from $-\infty$ to $+\infty$ over an interval of 180° .

Properties, and what they are for

$$\sin(-x) = -\sin x \cdot \cos(-x) = \cos x \cdot \tan(-x) = -\tan x$$

$$\sin^2x + \cos^2x = 1$$

The last is Pythagoras on the unit circle: the point $(\cos \theta, \sin \theta)$ is at distance 1 from the origin. Every trigonometric identity of Quarter IV is built from it.

QUADRANT	<i>SIN</i>	<i>COS</i>	<i>TAN</i>
I ($0\text{--}90^\circ$)	+	+	+
II ($90\text{--}180^\circ$)	+	−	−
III ($180\text{--}270^\circ$)	−	−	+
IV ($270\text{--}360^\circ$)	−	+	−

Periodic processes

$$y = A \sin(Bx) + C$$

A is the **amplitude** — half the distance from lowest to highest. C is the middle value.

The period is $\frac{360^\circ}{B}$.

PROCESS	PERIOD	MODEL
daylight hours in Tashkent	365 days	$12 + 3 \sin(\dots)$
tides	≈ 12.4 hours	$mean + amplitude \cdot \sin(\dots)$
mains alternating current	0.02 s	$220\sqrt{2} \sin(100\pi t)$
a Ferris wheel	one revolution	$centre\ height + radius \cdot \sin(\dots)$

02

Worked examples

EXAMPLE 1

State the period, amplitude and range of $y = 3 \sin(2x) + 1$.

1

Amplitude $|A| = 3$.

2

Period $\frac{360^\circ}{2} = 180^\circ$.

3

Middle value 1.

4

Range 1 ± 3 .

ANSWER

Period 180° , amplitude 3, range $[-2, 4]$

EXAMPLE 2 Without a calculator, give the sign of $\sin 200^\circ$, $\cos 200^\circ$ and $\tan 200^\circ$.

- ① 200° is in quadrant III.
- ② There sine and cosine are both negative.
- ③ Their quotient is positive.

ANSWER

$-$, $-$, $+$

EXAMPLE 3 Daylight in Tashkent varies from 9.3 h to 15.1 h. Find A and C.

- ① $C = \frac{15.1 + 9.3}{2} = 12.2$

The middle.

- ② $A = \frac{15.1 - 9.3}{2} = 2.9$

Half the swing.

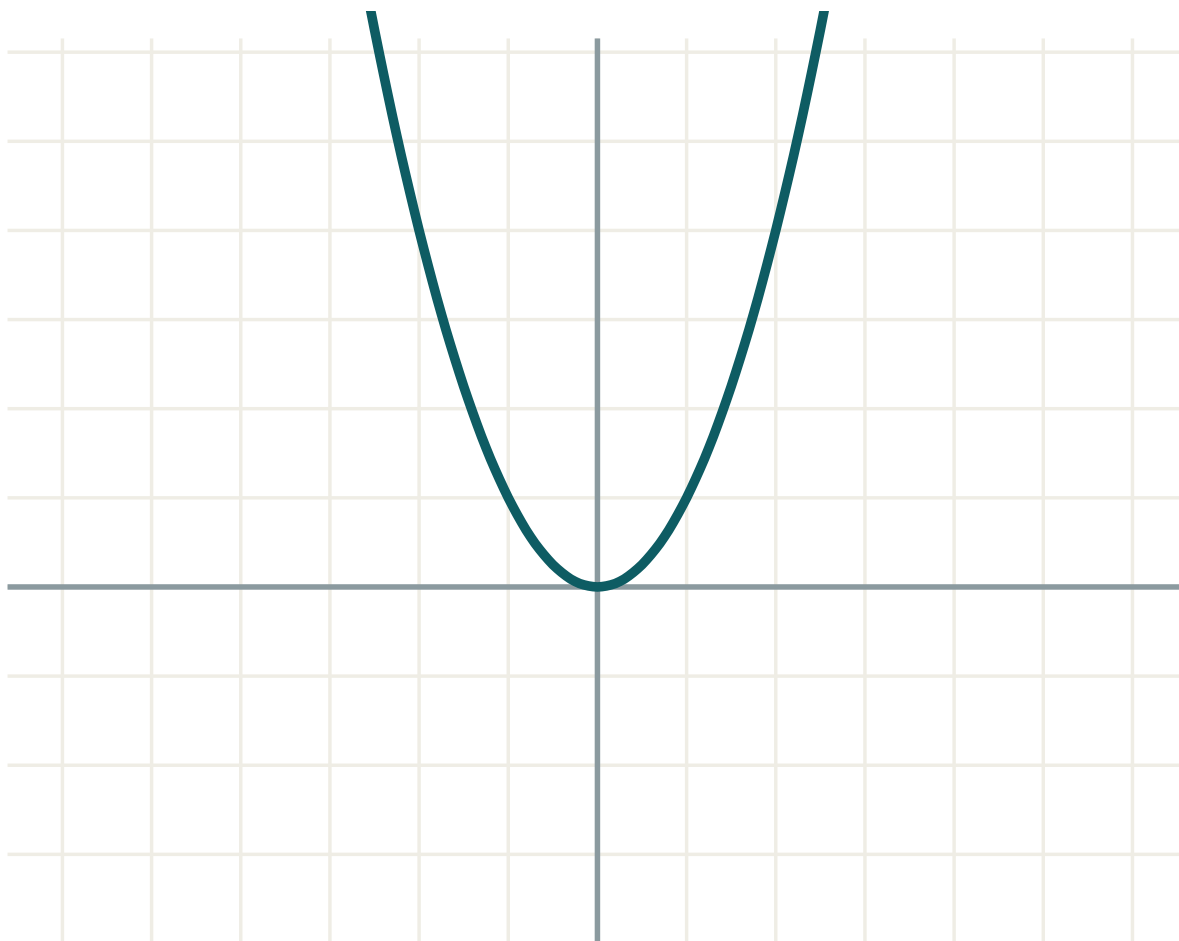
- ③ Period 365 days.

ANSWER

$A = 2.9$, $C = 12.2$ hours

03 Interactive model

Trace a point round the unit circle and plot its height against the angle.

A sine wave**a** 1**b** 0**c** 0equation **y = f(x)****04 Terminology**

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Trigonometric function	Trigonometrik funksiya	Тригонометрическая функция
Unit circle	Birlik aylana	Единичная окружность
Period	Davr	Период
Amplitude	Amplituda	Амплитуда
Zero of a function	Funksiya nuli	Нуль функции
Even function	Juft funksiya	Чётная функция
Odd function	Toq funksiya	Нечётная функция
Asymptote	Asimptota	Асимптота
Periodic process	Davriy jarayon	Периодический процесс

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The period of $\sin x$ is:

The period of $\tan x$ is:

$\cos(-x)$ equals:

The range of $\cos x$ is:

$\mathbb{R}$

$[-1, 1]$

$[0, 1]$

$x \neq 0$

The amplitude of $y = 3 \sin 2x$ is:

2

3

6

1.5

The period of $y = 3 \sin 2x$ is:

180°

360°

720°

120°

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Period of $\sin x$

ANSWER 360°

2. Period of $\tan x$

ANSWER 180°

3. Range of $\sin x$

ANSWER $[-1, 1]$

4. Zeros of $\sin x$

ANSWER $180^\circ k$

5. Is $\cos x$ even or odd?

ANSWER even

6. Is $\sin x$ even or odd?

ANSWER odd

7. $\sin 0^\circ$

ANSWER 0

Medium · the standard question

7 PROBLEMS

1. Period and amplitude of $y = 3 \sin 2x$

ANSWER $180^\circ, 3$

2. Range of $y = 3 \sin 2x + 1$

ANSWER $[-2, 4]$

3. Signs of $\sin$, $\cos$, $\tan$ at 200°

ANSWER $-, -, +$

4. Signs at 130°

ANSWER $+, -, -$

5. Period of $y = \cos(3x)$

ANSWER 120°

6. Amplitude of $y = -5 \cos x$

ANSWER 5

7. Where is $\tan x$ undefined?

ANSWER $x = 90^\circ + 180^\circ k$

Hard · stretch and reason

7 PROBLEMS

1. Daylight 9.3 to 15.1 h: find A, C

ANSWER 2.9, 12.2

2. A Ferris wheel of radius 20 m, centre 22 m up, one turn in 4 min: model the height

ANSWER $22 + 20 \sin(90t^\circ)$

3. Range of $y = 2 - 4 \cos x$

ANSWER $[-2, 6]$

4. Period of $y = \sin\left(\frac{x}{2}\right)$

ANSWER 720°

5. Solve $\sin x = 0$ on $[0^\circ, 720^\circ]$

ANSWER 0, 180, 360, 540, 720

6. Show $\sin^2 x + \cos^2 x = 1$ from the unit circle

ANSWER It is Pythagoras on the radius

7. Sketch $y = \tan x$ on $[-180^\circ, 180^\circ]$ with its asymptotes

ANSWER Asymptotes at $\pm 90^\circ$

07 Homework

Homework — 5 tasks

Every sketch must be labelled with the period and, for the tangent, the asymptotes.

- Sketch $y = \sin x$ and $y = \cos x$ on $[0^\circ, 720^\circ]$ on one set of axes.
- State the period, amplitude and range of $y = 4 \sin(3x) - 2$.
- Give the signs of the three functions at 110° , 250° and 320° .
- A tide varies from 1.2 m to 4.8 m with period 12.4 hours. Find the amplitude and the mean level.

5. Sketch $y = \tan x$ on $[-180^\circ, 180^\circ]$, marking the asymptotes and the zeros.

The values of arcsin a, arccos a, arctan a and arccot a

Undoing a trigonometric function — which needs a restricted domain, and therefore a convention.

LESSON 73–74

2 × 40 MIN

ALGEBRA 10, §4.2

P1 · 5.3

LESSON CLOCK

14'

22'

24'

20'

16'

8'

Why a restriction is needed	14 min
The four ranges	22 min
Standard values	24 min
One value or many?	20 min
Practice	16 min
Homework	8 min

LEARNING OBJECTIVES

- Explain why the trigonometric functions must be restricted to be invertible.
- State the principal-value ranges of the four inverse functions.
- Evaluate the inverse functions at the standard values.
- Distinguish “the” inverse value from all the solutions of an equation.

TEXTBOOK

National	pp. 313–324
Cambridge	Pure Mathematics 1, pp. 105–110
Workbook	P1 Exercise 5C

01

Explanation

Why a restriction is needed

$\sin x = 0.5$ has infinitely many solutions: $30^\circ, 150^\circ, 390^\circ, \dots$. So the sine is many-to-one, and by the rule of Grade 10 Quarter I it has **no** inverse — until its domain is cut down to a piece on which it is one-to-one.

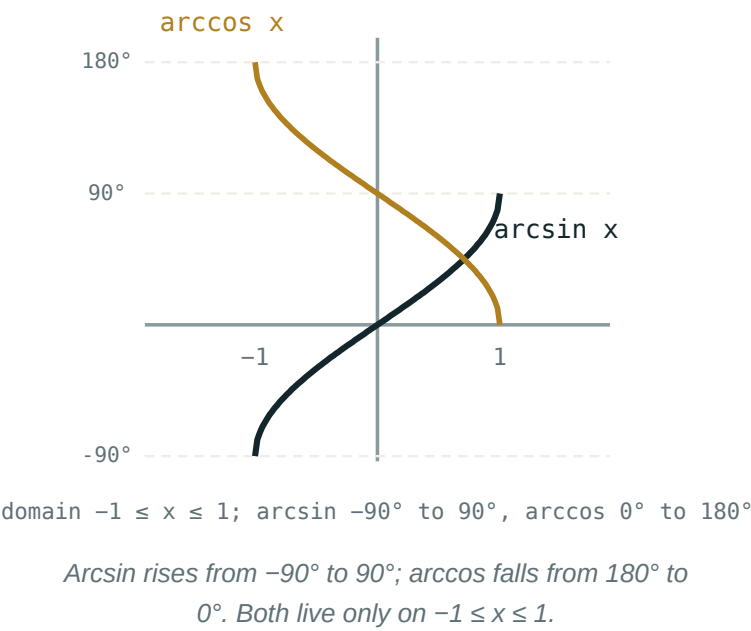
THE CONVENTION

Restrict the sine to $[-90^\circ, 90^\circ]$, where it increases from -1 to 1 exactly once. On that piece it is one-to-one, and its inverse is called **arcsin**.

The choice is a convention, but a universal one: every calculator and every textbook uses it, so an answer outside the range is simply wrong.

The four ranges

FUNCTION	DOMAIN	RANGE (PRINCIPAL VALUES)
$\arcsin a$	$-1 \leq a \leq 1$	$-90^\circ \leq y \leq 90^\circ$
$\arccos a$	$-1 \leq a \leq 1$	$0^\circ \leq y \leq 180^\circ$
$\arctan a$	$\mathbb{R}$	$-90^\circ < y < 90^\circ$
$\operatorname{arccot} a$	$\mathbb{R}$	$0^\circ < y < 180^\circ$



ARCCOS IS NEVER NEGATIVE

$\arccos(-0.5) = 120^\circ$, not -60° . The arcsine's range straddles zero; the arccosine's does not. Confusing the two is the standard error.

Standard values

A	$\text{ARCSIN } A$	$\text{ARCCOS } A$	$\text{ARCTAN } A$
0	0°	90°	0°
$\frac{1}{2}$	30°	60°	—
$\frac{\sqrt{2}}{2}$	45°	45°	—
$\frac{\sqrt{3}}{2}$	60°	30°	—
1	90°	0°	45°
$\sqrt{3}$	—	—	60°
$-\frac{1}{2}$	-30°	120°	—

Two identities are worth knowing, and both follow from the ranges:

$$\arcsin(-a) = -\arcsin a \quad \arccos(-a) = 180^\circ - \arccos a$$

$$\arcsin a + \arccos a = 90^\circ$$

One value or many?

TWO DIFFERENT QUESTIONS

“**Evaluate** $\arcsin 0.5$ ” has one answer: 30° . “**Solve** $\sin x = 0.5$ ” has infinitely many: $30^\circ + 360^\circ k$ and $150^\circ + 360^\circ k$. The first is a function value; the second is an equation.

Lesson 79–81 of Quarter IV is entirely about turning the first into the second. For now the distinction is the whole point: $\arcsin$ gives you **one** angle, chosen by convention, and the other solutions must be produced separately.

02 Worked examples

EXAMPLE 1 Evaluate $\arcsin(-\frac{\sqrt{2}}{2})$, $\arccos(-\frac{1}{2})$ and $\arctan(-1)$.

① $\arcsin \frac{\sqrt{2}}{2} = 45^\circ$, and arcsine is odd.
 -45°

② $\arccos \frac{1}{2} = 60^\circ$; use $180^\circ - 60^\circ$.
 120°

③ $\arctan 1 = 45^\circ$, and arctangent is odd.
 -45°

ANSWER

$-45^\circ, 120^\circ, -45^\circ$

EXAMPLE 2 Evaluate $\sin(\arccos 0.6)$.

① Let $\theta = \arccos 0.6$, so $\cos \theta = 0.6$ and $0 \leq \theta \leq 180^\circ$.

② $\sin^2 \theta = 1 - 0.36 = 0.64$

③ $\sin \theta = 0.8$

Positive, since θ is in the first two quadrants.

ANSWER

0.8

EXAMPLE 3 Why is $\arccos(-0.7)$ not -45.6° ?

- ① The range of $\arccos$ is $[0^\circ, 180^\circ]$.
- ② A negative value is impossible.
- ③ $\arccos(-0.7) = 180^\circ - \arccos 0.7 \approx 134.4^\circ$

ANSWER

It lies outside the range; the answer is $\approx 134.4^\circ$

03 **Interactive model**

Draw the restricted piece of each curve before evaluating anything.

Principal values*arcsin 1:* 0° 45° 90° 180° *arccos 0:* 0° 90° 180° -90° *arccos(-0.5):* -60° 60° 120° 240°

$\arcsin(-0.5)$:

-30°

30°

210°

330°

The range of $\arctan$ is:

$[0^\circ, 180^\circ]$

$(-90^\circ, 90^\circ)$

$[-90^\circ, 90^\circ]$

$\mathbb{R}$

$\arcsin a + \arccos a$ equals:

0°

90°

180°

it depends

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	ЎЗБЕКЧА	РУССКИЙ
Inverse trigonometric function	Teskari trigonometrik funksiya	Обратная тригонометрическая функция
Arcsine	Arksinus	Арксинус
Arccosine	Arkkosinus	Арккосинус
Arctangent	Arktangens	Арктангенс
Arccotangent	Arkkotangens	Арккотангенс
Principal value	Bosh qiymat	Главное значение
Restricted domain	Cheklangan soha	Ограниченная область
One-to-one	Biror-birga	Взаимно однозначное
Standard angle	Standart burchak	Табличный угол

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

A function needs a restricted domain to be inverted because it must be:

even

one-to-one

positive

periodic

The range of $\arcsin$ is:

$[0^\circ, 180^\circ]$

$[-90^\circ, 90^\circ]$

$[0^\circ, 90^\circ]$

$\mathbb{R}$

$\arccos(-a)$ equals:

$-\arccos a$

$180^\circ - \arccos a$

$\arccos a$

$90^\circ - \arccos a$

“Evaluate $\arcsin 0.5$ ” has:

no answer

one answer

two answers

infinitely many

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $\arcsin 0$

ANSWER 0°

2. $\arcsin 1$

ANSWER 90°

3. $\arccos 1$

ANSWER 0°

4. $\arccos 0$

ANSWER 90°

5. $\arctan 0$

ANSWER 0°

6. $\arctan 1$

ANSWER 45°

7. $\arcsin \frac{1}{2}$

ANSWER 30°

Medium · the standard question

7 PROBLEMS

1. $\arccos \frac{1}{2}$

ANSWER 60°

2. $\arcsin(-\frac{\sqrt{2}}{2})$

ANSWER -45°

3. $\arccos(-\frac{1}{2})$

ANSWER 120°

4. $\arctan(-1)$

ANSWER -45°

5. $\arctan \sqrt{3}$

ANSWER 60°

6. $\arcsin 0.6 + \arccos 0.6$

ANSWER 90°

7. $\arccos(-1)$

ANSWER 180°

Hard · stretch and reason

7 PROBLEMS

1. $\sin(\arccos 0.6)$

ANSWER 0.8

2. $\cos(\arcsin \frac{3}{5})$

ANSWER $\frac{4}{5}$

3. $\tan(\arcsin \frac{5}{13})$

ANSWER $\frac{5}{12}$

4. $\arccos(\cos 300^\circ)$

ANSWER 60°

5. $\arcsin(\sin 200^\circ)$

ANSWER -20°

6. Prove $\arcsin a + \arccos a = 90^\circ$

ANSWER Complementary angles in a right triangle

7. Why is $\arcsin(\sin 200^\circ) \neq 200^\circ$?

ANSWER 200° is outside the range of arcsin**07 Homework****Homework — 5 tasks**

Check every answer lies inside the correct range before writing it.

1. Evaluate $\arcsin \frac{\sqrt{3}}{2}$, $\arccos(-\frac{\sqrt{2}}{2})$, $\arctan(-\sqrt{3})$.

2. Evaluate $\sin(\arccos 0.8)$ and $\cos(\arcsin \frac{5}{13})$.

3. Evaluate $\arccos(\cos 250^\circ)$ and $\arcsin(\sin 160^\circ)$.
4. Explain in three sentences why the sine must be restricted before it can be inverted.
5. State the four principal-value ranges from memory and sketch the arcsine and arccosine curves.

The functions $y = \arcsin x$, $y = \arccos x$, $y = \arctan x$, $y = \operatorname{arccot} x$

The four inverse curves — their shapes, symmetries and asymptotes, obtained by one reflection.

LESSON 75–76

2 × 40 MIN

ALGEBRA 10, §4.3

P1 · 5.3

LESSON CLOCK

12'

24'

22'

22'

16'

8'

One reflection, four curves	12 min
Arcsin and arccos	24 min
Arctan and arccot	22 min
Transformations	22 min
Practice	16 min
Homework	8 min

LEARNING OBJECTIVES

- Sketch each inverse trigonometric function from the restricted original.
- State the domain, range, monotonicity and symmetry of each.
- Identify the asymptotes of arctan and arccot.
- Apply transformations to the inverse curves.

TEXTBOOK

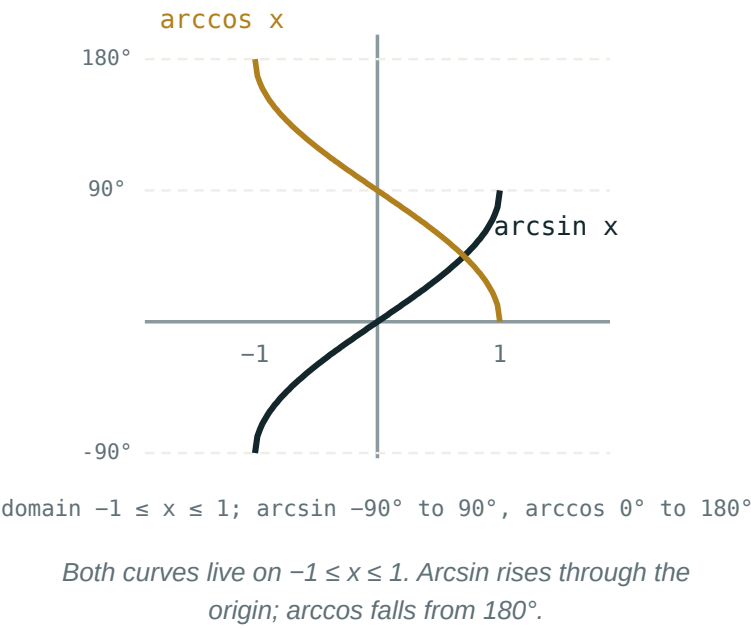
National	pp. 325–336
Cambridge	Pure Mathematics 1, pp. 105–110
Workbook	P1 Exercise 5C

01

Explanation

One reflection, four curves

Take the restricted branch of each trigonometric function and reflect it in $y = x$. The domain and range swap; increasing stays increasing; the x - and y -intercepts exchange.



EVERYTHING FOLLOWS FROM THE ORIGINAL

A vertical asymptote of the original becomes a **horizontal** asymptote of the inverse. A maximum becomes a right-hand endpoint. There is nothing new to learn — only to reflect.

Arcsin and arccos

	$Y = \text{ARCSIN } X$	$Y = \text{ARCCOS } X$
domain	$[-1, 1]$	$[-1, 1]$
range	$[-90^\circ, 90^\circ]$	$[0^\circ, 180^\circ]$
behaviour	increasing	decreasing
through	$(0, 0)$	$(0, 90^\circ)$
endpoints	$(-1, -90^\circ), (1, 90^\circ)$	$(-1, 180^\circ), (1, 0^\circ)$
symmetry	odd — half-turn about O	half-turn about $(0, 90^\circ)$

The relation $\arcsin x + \arccos x = 90^\circ$ says the two curves are reflections of each other in the horizontal line $y = 45^\circ$ — a fact visible at a glance in the figure.

Arctan and arccot

	$Y = \operatorname{ARCTAN} X$	$Y = \operatorname{ARCCOT} X$
domain	$\mathbb{R}$	$\mathbb{R}$
range	$(-90^\circ, 90^\circ)$	$(0^\circ, 180^\circ)$
behaviour	increasing	decreasing
through	$(0, 0)$	$(0, 90^\circ)$
asymptotes	$y = \pm 90^\circ$	$y = 0^\circ, y = 180^\circ$
symmetry	odd	half-turn about $(0, 90^\circ)$

THE ASYMPTOTES ARE HORIZONTAL, AND NEVER REACHED

$\arctan x$ climbs towards 90° for large x but never attains it — because $\tan 90^\circ$ does not exist. The range is an **open** interval, which is why the endpoints are written with round brackets.

Transformations

The rules of Quarter I apply unchanged:

CURVE	DOMAIN	RANGE
$y = \arcsin x$	$[-1, 1]$	$[-90^\circ, 90^\circ]$
$y = \arcsin(2x)$	$[-0.5, 0.5]$	$[-90^\circ, 90^\circ]$
$y = 2 \arcsin x$	$[-1, 1]$	$[-180^\circ, 180^\circ]$
$y = \arcsin(x - 1)$	$[0, 2]$	$[-90^\circ, 90^\circ]$
$y = \arctan x + 90^\circ$	$\mathbb{R}$	$(0^\circ, 180^\circ)$

The last row is worth noticing: $\arctan x + 90^\circ$ has exactly the range of $\operatorname{arccot} x$, and in fact $\operatorname{arccot} x = 90^\circ - \arctan x$ — the two are reflections in a horizontal line, as the arcsine and arccosine were.

02 Worked examples

EXAMPLE 1 State the domain and range of $y = \arccos(x - 2)$.

- ① Need $-1 \leq x - 2 \leq 1$.
- ② $1 \leq x \leq 3$
- ③ The range is unchanged by a horizontal shift.

ANSWER

Domain $[1, 3]$, range $[0^\circ, 180^\circ]$

EXAMPLE 2 State the domain and range of $y = 3 \arcsin(2x)$.

- ① $-1 \leq 2x \leq 1 \Rightarrow -0.5 \leq x \leq 0.5$
- ② The outer 3 stretches the range.
- ③ $3 \times [-90^\circ, 90^\circ]$

ANSWER

Domain $[-0.5, 0.5]$, range $[-270^\circ, 270^\circ]$

EXAMPLE 3 What are the asymptotes of $y = \arctan(x) - 30^\circ$?

① $\arctan x$ has asymptotes $y = \pm 90^\circ$.

② A vertical shift moves them by -30° .

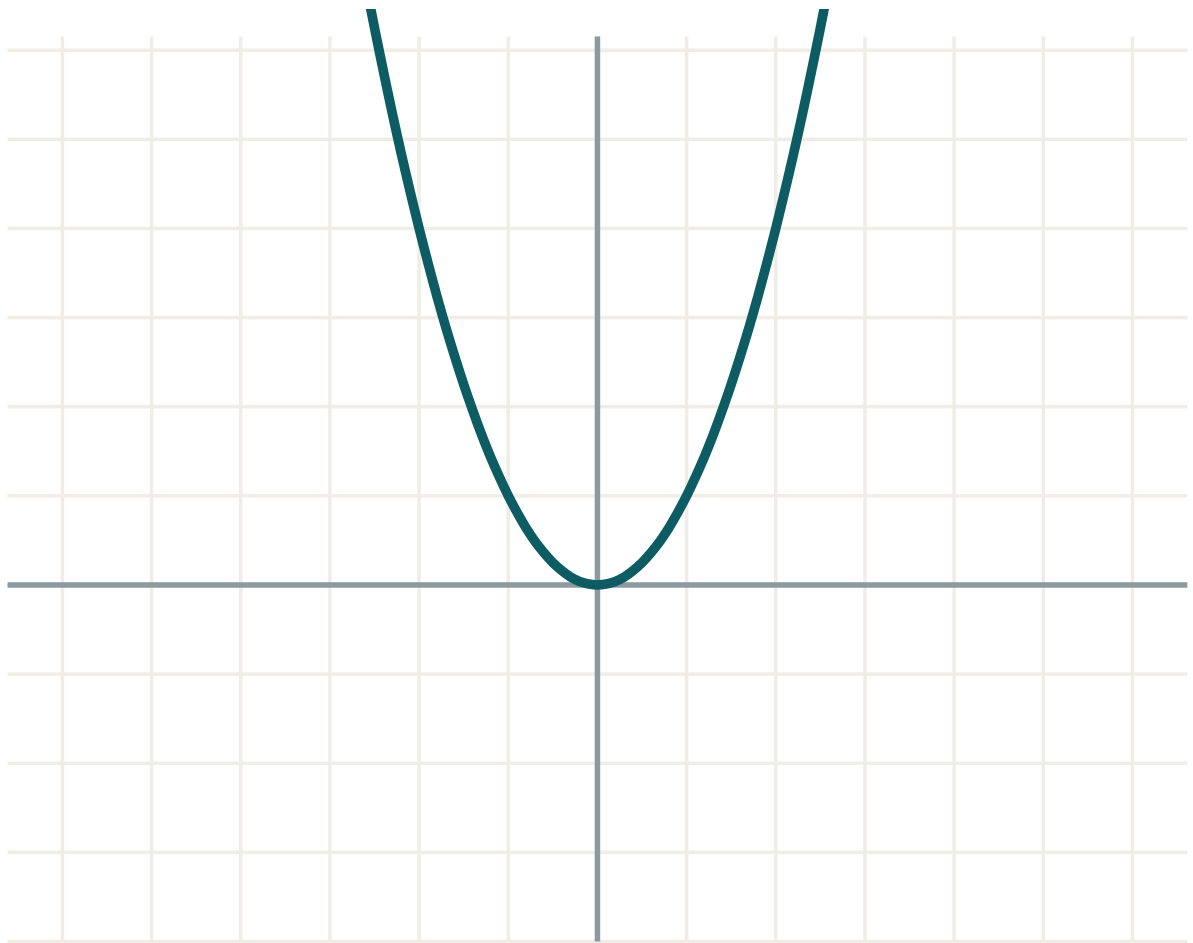
ANSWER

$$y = 60^\circ \text{ and } y = -120^\circ$$

03 **Interactive model**

Reflect the sine curve in $y = x$ with tracing paper — the arcsine appears exactly.

Reflecting a curve



a 1

b 0

c 0

equation $y = f(x)$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Inverse function	Teskari funksiya	Обратная функция
Reflection in $y = x$	$y = x$ ga nisbatan akslantirish	Симметрия относительно $y = x$
Restricted branch	Cheklangan tarmoq	Ограниченная ветвь
Horizontal asymptote	Gorizontal asimptota	Горизонтальная асимптота
Increasing function	O'suvchi funksiya	Возрастающая функция
Decreasing function	Kamayuvchi funksiya	Убывающая функция
Odd function	Toq funksiya	Нечётная функция
Centre of symmetry	Simmetriya markazi	Центр симметрии
Point of inflection	Egilish nuqtasi	Точка перегиба

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The domain of $\arcsin x$ is:

$\arccos x$ is:

The asymptotes of $\arctan x$ are:

The domain of $\arcsin(2x)$ is:

$[-1, 1]$

$[-0.5, 0.5]$

$[-2, 2]$

$\mathbb{R}$

The range of $2 \arccos x$ is:

$[0^\circ, 180^\circ]$

$[0^\circ, 360^\circ]$

$[-180^\circ, 180^\circ]$

$[0^\circ, 90^\circ]$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Domain of $\arcsin x$

ANSWER $[-1, 1]$

2. Range of $\arcsin x$

ANSWER $[-90^\circ, 90^\circ]$

3. Range of $\arccos x$

ANSWER $[0^\circ, 180^\circ]$

4. Range of $\arctan x$

ANSWER $(-90^\circ, 90^\circ)$

5. Is $\arcsin x$ increasing?

ANSWER yes

6. Is $\arccos x$ increasing?

ANSWER no

7. $\arccos x$ passes through

ANSWER $(0, 90^\circ)$

Medium · the standard question

7 PROBLEMS

1. Domain of $\arccos(x - 2)$

ANSWER $[1, 3]$

2. Domain of $\arcsin(2x)$

ANSWER $[-0.5, 0.5]$

3. Range of $3 \arcsin x$

ANSWER $[-270^\circ, 270^\circ]$

4. Range of $\arctan x + 90^\circ$

ANSWER $(0^\circ, 180^\circ)$

5. Asymptotes of $\arctan x - 30^\circ$

ANSWER $y = 60^\circ$ and $y = -120^\circ$

6. Domain of $\arcsin(x + 1)$

ANSWER $[-2, 0]$

7. Range of $-\arccos x$

ANSWER $[-180^\circ, 0^\circ]$

Hard · stretch and reason

7 PROBLEMS

1. Domain and range of $y = \arcsin(3x - 1)$

ANSWER $[0, \frac{2}{3}], [-90^\circ, 90^\circ]$

2. Domain and range of $y = 2 \arccos(\frac{x}{2}) - 90^\circ$

ANSWER $[-2, 2], [-90^\circ, 270^\circ]$

3. Show $\operatorname{arccot} x = 90^\circ - \arctan x$

ANSWER Complementary, with matching ranges

4. Sketch $y = \arcsin x$ and $y = \arccos x$ and mark the line of symmetry

ANSWER $y = 45^\circ$

5. Where do $\arcsin x$ and $\arccos x$ meet?

ANSWER $x = \frac{\sqrt{2}}{2}$, at 45°

6. Domain of $\arcsin(x^2)$

ANSWER $[-1, 1]$

7. Explain why the range of $\arctan$ is open at both ends

ANSWER $\tan(\pm 90^\circ)$ does not exist

07 Homework

Homework — 5 tasks

Every sketch needs the endpoints or the asymptotes marked.

- Sketch $y = \arcsin x$ and $y = \arccos x$ on one set of axes, with endpoints.
- Sketch $y = \arctan x$ with its asymptotes.
- State the domain and range of $y = \arccos(2x + 1)$ and $y = 2 \arcsin(x - 1)$.
- Find where $y = \arcsin x$ and $y = \arccos x$ intersect.

5. Show that $\operatorname{arccot} x = 90^\circ - \operatorname{arctan} x$ and check both ranges agree.

Project work — radians, arc length and sector area

Cambridge insert: the angle measure that makes every formula in calculus simpler, built and used in two lessons.

LESSON 77–78

2 × 40 MIN

ALGEBRA 10, §4.3 (EXTENSION)

P1 · 4.1–4.3

LESSON CLOCK

14'

18'

24'

20'

24'

8'

What a radian is	14 min
Converting	18 min
Arc and sector	24 min
The segment	20 min
The project	24 min
Homework	8 min

LEARNING OBJECTIVES

- Define the radian and convert between radians and degrees.
- Use $s = r\theta$ for arc length and $A = \frac{1}{2}r^2\theta$ for sector area.
- Find the area of a segment.
- Present a measured real object using radian formulas.

TEXTBOOK

National	pp. 337–342
Cambridge	Pure Mathematics 1, pp. 82–95
Workbook	P1 Exercise 4A–4C

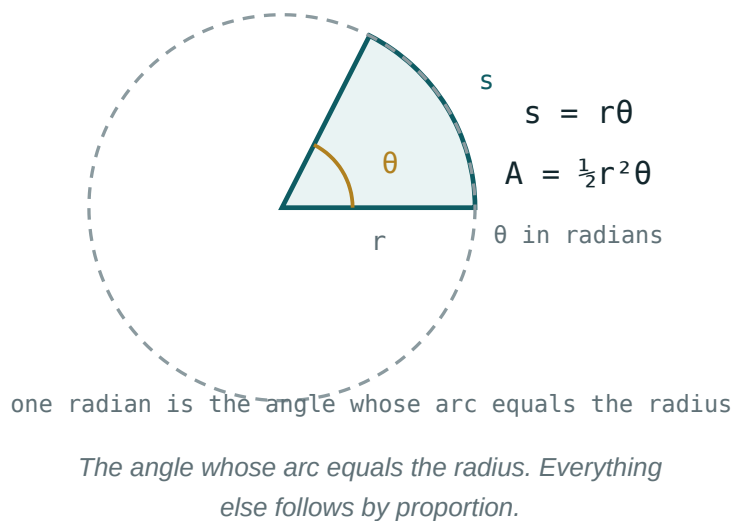
01

Explanation

What a radian is

DEFINITION

One **radian** is the angle at the centre of a circle subtended by an arc equal in length to the radius. It is a pure ratio $\frac{\text{arc}}{\text{radius}}$, so it has no units.



The whole circle has circumference $2\pi r$, so a full turn is 2π radians:

$$2\pi \text{ radians} = 360^\circ \quad \pi \text{ radians} = 180^\circ \quad 1 \text{ rad} \approx 57.3^\circ$$

Why bother. Degrees are an arbitrary Babylonian choice. Radians make the formulas below as simple as possible — and in Grade 11 they make $(\sin x)' = \cos x$ true, which it is not in degrees.

Converting

$$\text{degrees} \rightarrow \text{radians: multiply by } \frac{\pi}{180} \quad \text{radians} \rightarrow \text{degrees: multiply by } \frac{180}{\pi}$$

DEGREES	30°	45°	60°	90°	180°	270°
Radians	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	π	$\frac{3\pi}{2}$

SET THE CALCULATOR

Every wrong answer in this topic starts with the calculator in the wrong mode. $\sin 1$ is 0.0175 in degrees and 0.8415 in radians. Check the mode before every calculation.

Arc length and sector area

$$s = r\theta \quad A = \frac{1}{2}r^2\theta \quad (\theta \text{ in radians})$$

Both come from proportion: the sector is the fraction $\frac{\theta}{2\pi}$ of the whole circle, so its arc is that fraction of $2\pi r$ and its area that fraction of πr^2 .

IN DEGREES	IN RADIANS
$s = \frac{\theta}{360} \cdot 2\pi r$	$s = r\theta$
$A = \frac{\theta}{360} \cdot \pi r^2$	$A = \frac{1}{2} r^2 \theta$

The radian versions have no π and no 360 in them. That is the whole practical argument for radians.

The segment, and the project

A **segment** is the region between a chord and its arc — the sector minus the triangle:

$$A_{\text{segment}} = \frac{1}{2} r^2 \theta - \frac{1}{2} r^2 \sin \theta = \frac{1}{2} r^2 (\theta - \sin \theta)$$

THE PROJECT – TWO LESSONS, IN PAIRS

Find a real circular object or arc: a plate, a clock face, a fan, an arch of a madrasa, a wheel. Measure its radius and one angle. Then compute and present: the arc length, the sector area, the segment area, and one quantity of your own choosing.

One A3 sheet: a photograph or sketch with the measurements marked, the four calculations in radians, and one sentence on the accuracy of your measurement (Grade 11 will call these bounds).

Two pairs present in the last twenty minutes. The class asks one question each:

“How much would your answer change if your radius were 2 mm out?”

02 Worked examples

EXAMPLE 1 Convert 150° to radians and $\frac{5\pi}{6}$ to degrees.

$$(1) \quad 150 \times \frac{\pi}{180} = \frac{5\pi}{6}$$

$$(2) \quad \frac{5\pi}{6} \times \frac{180}{\pi} = 150^\circ$$

ANSWER

$$\frac{5\pi}{6} \text{ and } 150^\circ$$

EXAMPLE 2 A sector has radius 9 cm and angle 1.2 radians. Find its arc length and area.

$$(1) \quad s = 9 \times 1.2 = 10.8 \text{ cm.}$$

$$(2) \quad A = \frac{1}{2} \times 81 \times 1.2$$

$$(3) \quad = 48.6 \text{ cm}^2.$$

ANSWER

$$10.8 \text{ cm; } 48.6 \text{ cm}^2$$

EXAMPLE 3 Find the area of the segment cut off by a chord subtending $\frac{\pi}{3}$ at the centre of a circle of radius 6.

① Sector $\frac{1}{2} \times 36 \times \frac{\pi}{3} = 6\pi \approx 18.85$.

② Triangle $\frac{1}{2} \times 36 \times \sin \frac{\pi}{3} = 18 \times 0.866 \approx 15.59$.

③ Difference.

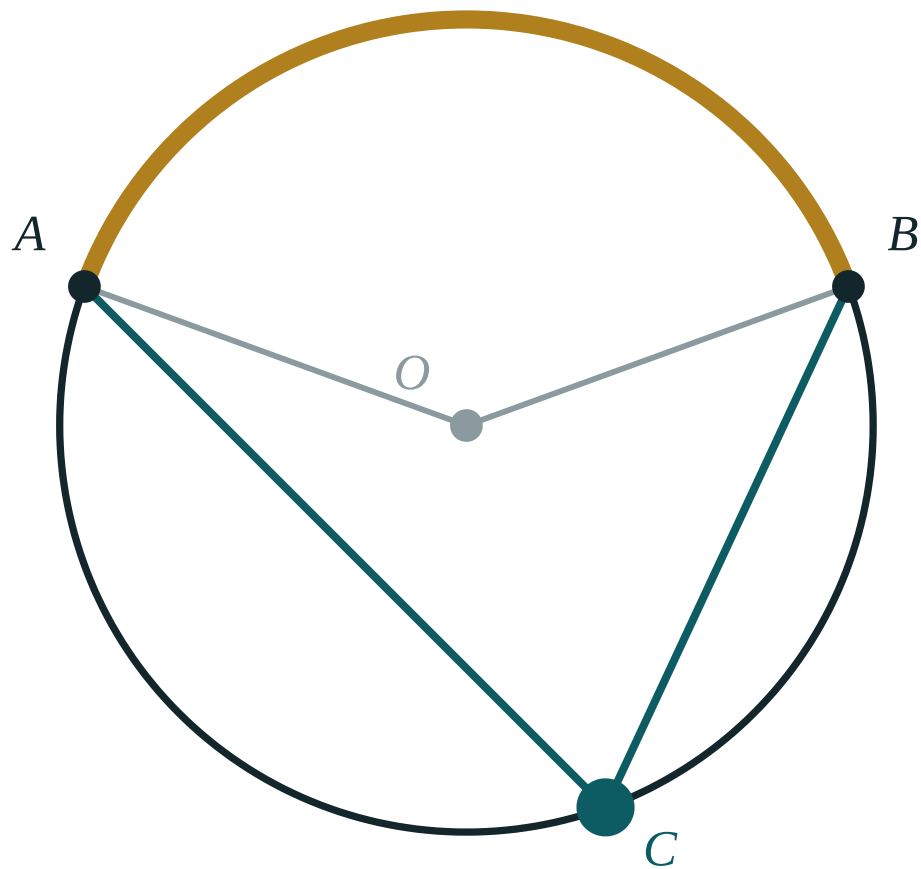
ANSWER

≈ 3.26 square units

03 Interactive model

Wrap a string of length r round the rim and mark the angle it subtends — that is one radian.

Arc, sector and segment

central $\angle AOB$ **140°** inscribed $\angle ACB$ **70°** ratio **2**arc AB **140°** **04 Terminology**

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Radian	Radian	Радиян
Arc length	Yoy uzunligi	Длина дуги
Sector	Sektor	Сектор
Segment	Segment	Сегмент
Central angle	Markaziy burchak	Центральный угол
Conversion	O'tkazish	Перевод
Chord	Vatar	Хорда
Subtend	Qarshi turmoq	Стягивать
Angular measure	Burchak o'lchovi	Угловая мера

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

π radians equals:

90°

180°

270°

360°

60° in radians is:

$\frac{\pi}{6}$

$\frac{\pi}{4}$

$\frac{\pi}{3}$

$\frac{\pi}{2}$

Arc length in radians is:

$$2\pi r$$

$$r\theta$$

$$\frac{1}{2}r^2\theta$$

$$\frac{\theta}{360}2\pi r$$

Sector area in radians is:

$$\pi r^2$$

$$r\theta$$

$$\frac{1}{2}r^2\theta$$

$$r^2\theta$$

A segment's area is:

sector plus triangle

sector minus triangle

half the sector

the whole circle

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Convert 90° to radians

ANSWER $\frac{\pi}{2}$

2. Convert 45° to radians

ANSWER $\frac{\pi}{4}$

3. Convert π to degrees

ANSWER 180°

4. Convert $\frac{\pi}{6}$ to degrees

ANSWER 30°

5. Arc length, $r = 5$, $\theta = 2$

ANSWER 10

6. Sector area, $r = 4$, $\theta = 1$

ANSWER 8

7. A full turn in radians

ANSWER 2π

Medium · the standard question

7 PROBLEMS

1. Convert 150° to radians

ANSWER $\frac{5\pi}{6}$

2. Convert $\frac{3\pi}{4}$ to degrees

ANSWER 135°

3. Arc length, $r = 9$, $\theta = 1.2$

ANSWER 10.8

4. Sector area, $r = 9$, $\theta = 1.2$

ANSWER 48.6

5. A sector of arc 15 and radius 6: its angle

ANSWER 2.5 rad

6. A sector of area 24 and radius 4: its angle

ANSWER 3 rad

7. Perimeter of a sector, $r = 5$, $\theta = 1.4$

ANSWER 17

Hard · stretch and reason

7 PROBLEMS

1. Segment area, $r = 6$, $\theta = \frac{\pi}{3}$

ANSWER ≈ 3.26

2. Segment area, $r = 10$, $\theta = 2$

ANSWER ≈ 54.5

3. A sector of perimeter 20 and radius 6: its area

ANSWER 24

4. A sector of area 50 and arc 20: find r and θ

ANSWER $r = 5$, $\theta = 4$

5. A chord of length 8 in a circle of radius 5: the segment area

ANSWER ≈ 7.9

6. A clock's minute hand is 12 cm. Distance the tip travels in 20 minutes

ANSWER $8\pi \approx 25.1$ cm

7. Two circles radius 5 with centres 6 apart: area of overlap

ANSWER ≈ 30.4

07 Homework

Homework — 5 tasks

Bring the finished project sheet to the first lesson of Quarter IV.

- Convert 120° , 225° and 40° to radians; convert $\frac{7\pi}{6}$ and 2.5 rad to degrees.
- A sector has radius 12 cm and angle 1.5 rad. Find its arc length, area and perimeter.
- Find the area of the segment cut off by a chord subtending $\frac{2\pi}{3}$ in a circle of radius 8.
- A sector of area 60 cm^2 has radius 10 cm. Find its angle in radians and in degrees.

5. Complete the project sheet: the object, the measurements, the four calculations and the accuracy sentence.